

The Incidence of No Tax on Tips

Dylan Balla-Elliott

Isaac Norwich*

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Using point-of-sale and payroll data, we estimate the direct tax savings, behavioral tipping response, and incidence of the 2025 No Tax on Tips (NTOT) provision. The average tipped worker directly saves \$531, but these savings are concentrated: 26% earn too little to benefit. Tip revenue rose 3.1% post-enactment as firms newly enabled tipping. Workers kept 72 cents of each new tip dollar, while firms lowered base pay by the remaining 28 cents. Overall, NTOT raised the average tipped worker's after-tax earnings by approximately 3.3%. The new tips are a raise for workers and, partly, a wage subsidy to firms.

*Balla-Elliott: University of Chicago, dbe@uchicago.edu. Norwich: University of Chicago, inorwich@uchicago.edu. We thank Nathaniel Hendren, Erik Hurst, Thibaut Lamadon, Matthew Notowidigdo, Ben Sprung-Keyser, Evan Rose, and Eric Zwick and seminar participants at the University of Chicago, the American Law and Economics Association Annual Meeting, and the Policy Impacts Annual Conference for helpful comments and suggestions. We also thank our data partner for providing access to the data used in this study. All analyses, findings, and conclusions are those of the authors and do not reflect the views of the data provider. This material is based on work supported by the National Science Foundation Graduate Research Fellowship under Grant No. DGE 1746045 and by the Policy Impacts Early Career Scholars Grant. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not reflect the views of the National Science Foundation or Policy Impacts.

On July 4, 2025, President Trump signed the No Tax on Tips (NTOT) provision into law, exempting the first \$25,000 of qualified tips from federal income taxation. The provision, part of the omnibus One Big Beautiful Bill Act (OBBBA), followed campaign pledges from the presidential candidates of both major US parties.¹ Several states have since enacted their own exemptions, and still more states are debating similar measures.

Despite this widespread policy interest, we have little empirical evidence of how tipped workers and the firms that employ them respond to this tax incentive. By advantaging tip earnings over wage earnings, NTOT incentivizes firms and workers to substitute from wages toward tips. This margin evidently concerns policymakers: To limit outright reclassification, the Treasury restricted the deduction to a list of 71 occupations that “customarily and regularly” receive tips, from waiters and bartenders to pet caretakers and water taxi operators (IRS, 2026). Substitution between wage and tip pay is also the heart of a longer-running debate over whether tips represent a raise for workers or a wage subsidy to firms.

In this paper, we estimate the effects of NTOT on tax savings and tipping behavior, and use these effects to recover the economic incidence of new tips on workers and firms. We find that NTOT raised the average tipped worker’s earnings net of federal income taxes by approximately 3.3%, through both direct tax savings and tips induced by the policy. Workers keep the majority of these new tips as a raise, though they are also a partial wage subsidy to firms.

The direct tax savings average \$531 per tipped worker but are highly concentrated: 26% of tipped workers earn too little to benefit because their total earnings fall below the standard deduction. To overcome the usual challenges in measuring tip income,² we measure tips in administrative payroll records and validate them against POS revenue. Using this novel data, we find that the tax savings for workers with bottom-quintile earnings are larger than pre-implementation projections (Gleckman, 2024; Council of Economic Advisers, 2025; The Budget Lab, 2024).

These direct estimates assume that tip revenue stays at its pre-policy level, but we find instead that NTOT increased tip revenue by causing some firms to newly enable tipping. Tip revenue rose 3.1% in the quarter following enactment, driven entirely by these firms. For the average tipped worker, these induced tips added \$193 per year to the direct tax relief.

However, these induced tips need not increase earnings one-to-one: if base pay adjusts, part of each tip dollar becomes a wage subsidy to firms. We use variation in the timing and intensity of firms’ adoption of tipping to estimate the effect of an extra dollar of induced tips on base pay and total earnings. Firms that newly enabled tipping clawed back about 28 cents of each induced tip dollar through lower base pay; workers kept the remaining 72 cents.³ Firms’ partial capture of the induced

¹Public Law 119-21. The deduction was retroactive for 2025 income and is scheduled to expire after 2028.

²Basker et al. (2024) estimate that tax data understate tip earnings by billions of dollars annually; Basker et al. (2026) show that under-reporting extends throughout the distribution, not just near reporting thresholds.

³*Clawback* refers to economic, not statutory, incidence. Tips are mechanically paid to workers in full, but if tip earnings make workers willing to accept lower base pay, part of the benefit accrues to firms. Our estimates do not imply that tips are withheld from workers.

tips parallels the effect of the earned income tax credit (EITC): Rothstein (2010) estimates that roughly 36 cents of each EITC dollar is absorbed by cuts to base pay.

Combining the tax savings, the induced tips, and the clawback, NTOT’s marginal value of public funds (MVPF) is at least 1 and could be as high as 1.40. Where the MVPF falls in this range depends on customers’ welfare gain from tipping.⁴ If the induced tips are a *pure transfer*—each tip dollar costs the customer a full dollar—then net willingness to pay (WTP) equals the government’s net cost, yielding an MVPF of exactly 1. However, tipping may instead generate surplus. In the limiting *tipping surplus* case—each tip dollar spent yields the customer a full dollar in welfare gain—induced tips create new surplus.⁵

In addition to providing the first empirical analysis of NTOT, we also contribute to the literature on the economics of tipping (Azar, 2020; Wessels, 1997). The existing empirical estimates of tip incidence come from gig platforms. When Uber enabled in-app tipping, total driver earnings changed little because driver entry and utilization adjusted (Chandar et al., 2024a; Cook et al., 2021; Hall et al., 2023). In contrast, we find that outside of the gig economy, workers keep the majority of new tips. This is consistent with labor market frictions determining economic incidence. In a perfectly competitive market such as Uber, competition dissipates the gains entirely; however, in a frictional labor market, wages adjust only partially. Our incidence exercise is closest in spirit to Jones (2016) and Talesara and Wickard (2025), who use tax data and state tipped-minimum-wage variation to show that increases in wage pay crowd out tips; Talesara and Wickard (2025) find that a 10% increase in the tipped minimum wage lowers tips by 4%, fully offsetting the gain in base pay. We estimate substitution in the opposite direction, asking how induced tips crowd out base wages.

1 Setting

The NTOT policy tax-advantages qualified tips over wages; this advantage incentivizes firms and workers to substitute compensation from wages toward tips,⁶ because shifting a dollar of compensation from base wages (fully taxable) to qualified tips (deductible from federal taxable income) lowers a worker’s tax bill. Because the policy, enacted in July 2025, was retroactive to the start of 2025, this incentive operated from the moment of enactment. Appendix A provides additional detail about the policy.

Tip income is largely unobserved in administrative data and is subject to systematic underreporting in surveys. Tax return data from W-2 forms capture only reported tips, missing substantial unreported amounts. Basker et al. (2024) estimate that \$8 billion in tips paid annually at full-service, single-location restaurants from 2005 to 2018 were not captured in tax data. The tip income data

⁴Azar (2020, p217) writes, “It is extremely difficult to explain tipping behavior from the perspective of a purely self-interested consumer.”

⁵For example, in models where tipping depends on social norms and other nonstandard psychological considerations, tipping can increase social surplus (Azar, 2005).

⁶The deduction applies only to federal income taxes, not payroll taxes, and phases out at higher income levels.

from the Survey of Income and Program Participation (SIPP) and Current Population Survey (CPS) rely on self-reports, which are subject to recall error or strategic underreporting.

In contrast, our administrative POS and payroll data from a large business technology platform allow us to observe each tip dollar twice: when it is collected as a payment from the customer and when it is distributed—in whole or part—to the worker as earnings. These directly observed data have much less scope for misreporting than standard survey data. In a robustness exercise, we use only the tipped earnings we can directly validate in the transaction data. Appendix C.3 presents these estimates.

Our data cover small firms, predominantly in food service with substantial retail representation, across all states from 2023 to 2025. The transaction records contain itemized receipts with tip amounts and tipping availability. The payroll records detail base wages, hours, overtime, and tip distributions.

From these broader data, we construct three analysis samples, one for each step of the analysis. We use 2024 payroll data for roughly 172,000 tipped workers at 13,000 firms to measure the direct effect (Section 2), a weekly firm-level transaction panel for approximately 5,700 high-volume firms across 2023–2025 to measure the tipping response (Section 3), and a worker-level panel around firms’ adoption of tipping to measure the incidence of the induced tips (Section 4). Appendix B further details the data source and its structure.

2 Direct Effect: Tax Savings from Tip Exemption

We use 2024 payroll data to measure the direct tax savings from NTOT in the absence of behavioral responses. The savings average \$531 per tipped worker but are highly concentrated: The top 20% capture 51% of the total, while 26% of tipped workers earn too little to benefit.

2.1 Sample and Tax Calculation

The sample contains roughly 172,000 tipped workers at 13,000 firms. A tipped worker is any worker with any tip earnings in 2024. For workers observed for fewer than 12 months, we impute annual earnings via donor matching: We find full-year workers who resemble them in location, industry, and monthly earnings patterns. The result is our combined sample. This approach ensures the sample represents the range of tipped workers in our data, not just those with stable year-long employment. Appendix C.1 details the matching procedure and sample composition.

Four in five workers in our sample are employed in food and beverage firms. Average monthly earnings are \$1,970, corresponding to an average of 90 hours’ pay at roughly \$22 an hour. The firms in our sample are small: 82.4% have fewer than 15 workers and 64% fewer than 10; approximately three-quarters of workers are at firms with fewer than 20. Appendix C.2 presents the full distribution of worker and firm characteristics.

To estimate tax savings, we calculate federal income tax liability for 2024 minus the counterfactual liability under NTOT. We make three assumptions. First, we assume that the observed yearly income for 2024 in our combined sample is equal to an individual’s total income reported on her taxes. Second, we assume that all the individuals in our sample are single filers.⁷ Third, when calculating modified adjusted gross income, we assume that she takes no deductions other than the standard deduction. These assumptions are reasonable given the types of workers in our sample and yield lower-bound estimates of the tax savings under NTOT.⁸

2.2 Results

We use the distribution of tip and wage earnings to estimate the direct effects of the NTOT tip income deduction on federal tax liability.

2.2.1 The tip share of earnings varies widely across tipped workers

The tipped workers in our sample vary widely in how much of their compensation comes from tips. Median yearly income is \$20,950, with 90% of the workers earning less than \$42,135, so our estimates are informative about the lower end of the earnings distribution. The share of annual earnings from tips averages 26%, with a long right tail: Workers in the top 10% of this distribution earn more than 46% of their income from tips, but those in the bottom 10% earn 7% or less. Appendix C.2 presents both distributions.

2.2.2 Tax savings exceed \$758 for the 75th percentile but are zero for 26%

Figure 1 reports the distribution of direct tax savings for our sample of tipped workers. There is considerable dispersion in the tax reduction. We estimate direct tax savings from NTOT of \$531 for the average individual. The top 25% save more than \$758 for the year, and the top 20% of workers by income account for 51% of all the tax savings from the NTOT provision. Meanwhile, 26% see no tax savings at all because their total earnings are below the standard deduction. The small spike in the tax savings distribution at \$3,000 reflects workers who take the maximum NTOT deduction of \$25,000 at the 12% marginal tax rate. The average tax savings are roughly 1.7% of income. Appendix C.2 shows the distribution of savings as a share of earnings and the concentration curve.

Among the 74% of tipped workers with positive tax savings, the average benefit is \$722 (median: \$529), which represents 2.25% of average earnings. The workers with positive savings earn slightly more on average than the full sample (\$29,054 vs. \$24,153), which reflects that those with income

⁷Or, equivalently, that they are married to someone in the same tax bracket and with no deductible tip earnings.

⁸If workers had additional income outside our data that pushed them into higher tax brackets, we would underestimate their tax savings. The single-filer assumption is also conservative. Under the US tax code in 2024, the income bands for married-filing-jointly filers were twice those for single filers up to \$243,726, so a worker married to someone with equal earnings would face the same marginal tax rate, which would make our estimates exact. A worker married to someone earning more would face higher marginal rates, so we would underestimate her tax savings. The standard-deduction assumption is well justified: 95.2% of tax filers with adjusted gross income (AGI) between \$1 and \$100,000 took the standard deduction in 2022 (IRS, Statistics of Income Division, 2022).

below the standard deduction threshold receive no benefit from the tip income deduction. We report results conditional on positive benefits to draw comparisons with prior estimates, which similarly condition on workers benefiting from the policy.

2.3 Discussion: Comparison with Other Estimates

Our results are qualitatively similar to existing estimates, though on the whole we find larger tax savings. Most of the direct benefits accrue to the workers with the highest tip earnings. A long tail of workers has positive but small gains. Quantitatively, our estimates of tax savings across the income distribution are larger than the projections produced by The Budget Lab at Yale (TBL), which is the study most similar to ours in its aims. Under their assumptions, our estimates of the bottom-quintile savings are 2.6 times theirs. See Appendix C.4 for further discussion of the technical differences between this study and the existing policy projections.

3 Behavioral Response: Changes in Tipping After Policy Passage

Tipping behavior responded to NTOT within weeks of the provision’s standalone Senate passage. The share of tippable transactions rose from 92% to 96%; net of firm-specific seasonality, the increase was 3–5 percentage points (pp). Tip revenue rose 3.1% in the quarter following enactment, driven by firms that newly enabled tipping after the policy passed; the firms already accepting tips at baseline adjusted very little. For the average tipped worker, this added \$193 per year to the direct tax relief.

3.1 Data and Measurement

We aggregate transaction records to form a weekly firm-level panel spanning January 2023–December 2025. Our main analytic sample contains firms active every month from January 2023 through December 2024 with roughly 1,000 or more transactions per week—some 5,700 firms that average \$57,900 in weekly gross transaction volume (revenue plus sales tax and tips). Tip revenue and the tip-to-revenue ratio are measured across all transactions; the share of tippable transactions is measured among card payments only.⁹ Appendix D.1 details sample construction.

The IRS made no changes to 2025 withholding tables, W-2 forms, or 1099 forms. Workers saw the deduction only when filing their 2025 tax returns in early 2026. The behavioral response we measure for 2025 therefore reflects salience and expectations rather than realized changes in paycheck-level disposable income.

⁹The platform records whether the customer was prompted to tip for all card payments, which correspond to the majority of revenue; cash payments account for only 9% of the revenue in our sample. Cash tips may be incompletely captured but are a small and declining share of tip income among the types of firms in our sample (Basker et al., 2024).

3.2 Research Design

The weekly series show strong seasonality, so the raw changes around the policy’s rollout do not identify its effect. Our identification strategy thus uses the difference between the changes since May 20, 2025, and the changes at the same firm over the same period in previous years. We call this a calendar-time event study, relying on a variation of the common trends assumption that we dub calendar-time common trends (CTCT).

The familiar difference-in-differences (DID) common trends assumption is that the change in the untreated potential outcome over time is the same in the untreated and treated groups. The CTCT assumption is that the change in the untreated potential outcome from one week *of the calendar year* to another is the same in the pre- and post-policy years.

Assumption 1 (CTCT). *Common Trends in Calendar Time with Unobserved Heterogeneity*

For every firm j , every calendar year $y(t)$, and every calendar week $k(t) \in \{1, \dots, 52\}$, the untreated potential outcome $Y_{jt}(0)$ satisfies

$$\mathbb{E}[Y_{js}(0) - Y_{jt}(0) \mid y(s) = y(t)] = \gamma_{j,k(s)} - \gamma_{j,k(t)} \quad \text{for all } t, s \text{ with } y(s) = y(t) \quad (1)$$

where $\gamma_{j,k(t)}$ is the firm-specific calendar-time trend (i.e., seasonality). The identifying assumption is that the calendar-time trends for a given firm j are the same across calendar years.

The CTCT assumption is testable: With three years of data, the model is overidentified. We can estimate the same design using 2024 as a placebo “policy year” and 2023 as a control year—an approach analogous to using pre-treatment periods to test common trends in the canonical event study. This exercise shows no systematic increase in tipping behavior around the analogous 2024 date, so calendar-year patterns do not drive the estimates that follow. Appendix D.3 presents the placebo estimates.

3.3 Results

Under NTOT, the share of tippable transactions rose by 3–5 pp and tip revenue by about \$100 per firm-week, while total revenue stayed flat. For the average tipped worker, the induced tips added \$193 per year.

3.3.1 The share of tippable transactions jumps from 92% to 96% at passage

The raw weekly series broke from their 2023–2024 paths in the weeks between the standalone NTOT Act’s Senate passage and the OBBBA’s signing: The share of tippable transactions jumped from 92% to 96% and remained at the elevated level (Figure 2a). Tip revenue diverged from its 2023–2024 path around May 20 (Figure 2b). The tip-to-revenue ratio—normally declining over this stretch of the year—held at or slightly above its 2023–2024 path (Figure 2c).

3.3.2 Empirical specification

Let $\ell(t)$ denote event time, the number of weeks between week t and a fixed calendar date within year $y(t)$, where $\ell(t) = 0$ corresponds to the week of the event.

We implement our CTCT design with the following estimation equation:

$$Y_{jt} = \alpha_{j,y(t)} + \gamma_{j,k(t)} + \sum_{\ell \notin \{-4, \dots, -1\}} \beta_{\ell} \mathbb{1}\{\ell(t) = \ell \text{ \& } y(t) = 2025\} + \varepsilon_{jt} \quad (2)$$

where Y_{jt} is the outcome for firm j in period t , $\alpha_{j,y(t)}$ are firm–year $y(t)$ fixed effects, $\gamma_{j,k(t)}$ are firm-specific calendar-week trends, and β_{ℓ} are the event-time coefficients of interest. The plot of β_{ℓ} presents the difference in outcomes between 2025 and 2023–2024 for each event-time bin relative to the four weeks prior to the event date.¹⁰ We compute Driscoll and Kraay (1998) standard errors, which accommodate both cross-sectional dependence across firms within each week and serial correlation within firms over time.¹¹

3.3.3 Tipping rises at firms; sales do not

After NTOT’s passage, the share of tippable transactions rose 3–5 pp, tip revenue rose about \$100 per firm–week, and the tip-to-revenue ratio rose roughly 0.15 pp, with statistically null pretrends. Figure 3 plots the β_{ℓ} coefficients from Equation (2) for these three outcomes, relative to the four weeks before May 20. The tip revenue estimates are noisy for the weeks early in the year, and we cannot reject that tips were below the calendar-time trend early in the year. On average, firms enabled tipping on a larger share of their transactions, which increased tip revenue. Meanwhile, gross transaction volume and revenue remained flat (Appendix D.3).

3.3.4 Firms newly enabling tipping drive the response

The aggregate effects are driven by firms that start 2025 with essentially no tipping. We classify firm–years by the share of transactions (weighted by revenue) that are tippable in the first four weeks of each year. We create two subsamples consisting of firm–years with shares of 0–5% (“near zero”) and 95–100% (“near one”).¹² These groups differ markedly in composition and baseline tipping behavior. Firm–years in the near-zero group average only \$44 in weekly tip revenue with a 24.8% share of tippable transactions, and those in the near-one group average \$1,113 in weekly tip revenue and a 98.6% share. The average transaction has a substantially higher bill size in the near-zero sample (\$33.51) than in the near-one group (\$18.70), reflecting the retail-heavy composition

¹⁰For the purpose of this analysis, the event date is May 20, when the standalone NTOT bill first passed the Senate. This is when salience around NTOT’s passage sharply increased.

¹¹We prefer this approach to standard firm-level clustering, which allows arbitrary within-firm correlation across the full panel and is overly permissive given that the residual autocorrelation decays to zero within 8 weeks. We set the lag length to 8 weeks, the bandwidth chosen by the Andrews (1991) automatic procedure.

¹²These subsamples are built using the full firm roster rather than the top-decile analysis sample; see Appendix D.1 for details.

of the former (Appendix D.1).

Figure 4 presents results for firms that start “near zero” and shows that tip revenue rose by over \$150 per week and the share of tippable transactions rose by approximately 80 pp. In contrast, we find no evidence of adjustment in the “near one” group, consistent with these firm-years already operating near full tipping saturation.

Appendix D.2 presents the near-zero and near-one panels and a sectoral decomposition of the results.

3.4 Discussion: Earnings Gains for the Average Tipped Worker

To translate these firm-level estimates into per-worker earnings changes, we assume that each existing worker’s tip income rose by the same percentage as firm tip revenue. In the quarter following July 4, tip revenue per firm increased by roughly \$100 per week, or 3.1% of pre-period average tip revenue. A 3.1% increase in tip income for the average tipped worker (earning \$6,215 in tips annually) corresponds to \$193 per year, or 0.8% of average gross earnings (\$24,153).

4 Incidence: Who Keeps the Induced Tips?

Firms that newly enabled tipping clawed back about 28 cents of every dollar of the NTOT-induced tips through lower base pay; workers kept the remaining 72 cents. Tips are mechanically distributed to workers, but nominal incidence is not economic incidence—as borne out by the research on EITC payments, which are captured partly by firms (Rothstein, 2010), and on the cost of payroll taxes remitted by firms, which may be borne fully by workers (Gruber, 1997) or fully by firms (Saez et al., 2019). We use variation in the timing and intensity of tipping adoption to estimate the effect of an extra dollar of induced tips on base pay and total earnings. Our confidence intervals exclude the two extremes: no clawback and full clawback.

4.1 Research Design

The clawback is identified from the timing of adoption. We assume that, conditional on a firm’s eventual tipping intensity, the precise time it turns tipping on is unrelated to its wage setting. Firms enabled tipping at different points in 2025; under common trends between early and later adopters, adoption timing identifies average effects on tips and on base pay.¹³

To increase power, we also use variation in the intensity of the induced increase in tips. Each adopter’s *dose* is its average tip income per hour over its first three full months of tipping. For each month, we would like to compare workers at high-dose firms that have adopted with workers at high-dose firms that have not yet adopted, low-dose adopters with low-dose not-yet-adopters, and so on across the dose distribution.

¹³Appendix E reports the unconditional adoption event study; the combined rate shows positive pre-adoption coefficients consistent with heterogeneity, motivating the dose-response design.

Under this common trends assumption, the observed contrast between adopters and not-yet-adopters at each dose identifies the causal effect of adopting at that dose:

$$\mathbb{E}[\Delta Y_{it} \mid D_i = d, A_{it} = 1] - \mathbb{E}[\Delta Y_{it} \mid D_i = d, A_{it} = 0] = \mathbb{E}[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d] \quad (3)$$

where Y_{it} is the base-wage rate or the combined hourly rate (base wage plus tips per hour) of worker i in month t , ΔY_{it} is its change relative to the rate in the pre-adoption period, D_i is the firm's dose, and A_{it} is an indicator for whether the firm has adopted tipping.¹⁴

The parameter of interest is the effect of adopting tipping at dose d , per dollar of dose,

$$\gamma_d \equiv \frac{\mathbb{E}[\Delta Y_{it}(d) - \Delta Y_{it}(0) \mid D_i = d]}{d} \quad (4)$$

The dose does not predict pre-adoption changes in either outcome (Appendix E).

4.2 Estimation and Results

4.2.1 Sample

The sample is established firms whose adoption plausibly responded to NTOT: firms with at least 24 months of payroll data in 2023–2025 that collected no tips in 2024 and began tipping between March and August 2025, when the policy's salience was highest (Appendix A). These restrictions exclude new entrants that opened with tipping. Firms that adopted tipping in September and October of 2025 enter as not-yet-treated controls.

The estimation is at the worker level. Each worker must be observed in every month from relative month -3 to 2 around the firm's adoption, and months outside this window enter the estimation when observed. The sample contains 689 workers at 156 firms.¹⁵

4.2.2 Estimating equation

The regression pools variation across adoption timing and across doses. We implement the comparison in Equation (3) as a stacked DID (Balla-Elliott and Norwich, 2025; Deshpande and Li, 2019) with the dose defined at the firm level:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \mathbb{1}\{k_{it} = k\} + \sum_{k \neq -1} \beta_k D_{j(i)} \mathbb{1}\{k_{it} = k, D_{j(i)} > 0\} + \varepsilon_{it} \quad (5)$$

where α_i and λ_t are worker and month fixed effects, k_{it} is worker i 's month relative to her firm's adoption month, δ_k is the adoption effect common to all adopters in relative month k , and β_k , the coefficient of interest, is on the interaction of the dose with adoption. Estimating a single coefficient

¹⁴Appendix E shows that the common trends assumption can be relaxed to hold only conditional on the dose; the identifying variation is still adoption timing.

¹⁵Appendix E details sample construction, the treatment of adopters whose measured dose is zero, and the estimator.

per relative month imposes that the response is constant in the dose.¹⁶ The restriction lets us compare workers not only within a dose level but also across doses.

We measure all variables in levels rather than logs so that the coefficients are in dollar units, as cents per dollar of dose. A percentage specification would divide a constant dollar offset by each firm’s wage level, disguising this offset as incomplete and heterogeneous pass-through in percentages (Sangani, 2026); using levels rather than logs also avoids artifacts arising from different bases.

4.2.3 Results

Firms clawed back 28 cents of every dollar of induced tips through base-wage reductions (SE 3.7, 95% CI [21, 35] cents), and workers retained the remaining 72 cents as a net compensation gain. Figure 5 shows the event-study coefficients for the base-wage rate and the combined hourly rate: Following adoption, each dollar of dose lowered the base-wage rate by 25.4 cents and raised the combined rate by 64.8 cents. The sum of these two responses is the flow of induced tips that a dollar of dose generates; dividing the base-wage decline by this flow converts it into the clawback share, in cents per dollar of induced tips.^{17,18} This flow need not equal one dollar. The dose is a firm-level average (total tips divided by total hours), but tips are distributed unevenly across workers within a firm. The worker-level regression coefficients therefore capture the average worker’s response per dollar of firm-level dose, which can differ from one dollar.

4.3 Discussion: Workers Keep the Majority of Each Induced Tip Dollar

Workers kept the majority of each dollar of induced tips: Of the \$193 in annual induced tips for the average tipped worker, firms clawed back roughly \$54, and workers retained the remaining \$139. Under full offset, base pay would fall one-for-one with new tips, and the clawback would be 100 cents on the dollar. Because tip income is no longer subject to federal income tax though it still bears payroll tax, however, even full offset increases workers’ after-tax income; to keep after-tax compensation unchanged, firms would need to claw back 115 cents on the dollar for a worker in the 12% bracket. Our estimates decisively reject both full clawback and no clawback: Firms and workers shared the benefits of the induced tips, though the majority of the gain accrued to tipped workers. The estimate captures the incidence of induced tip income at newly tipping firms, not the reclassification of existing wages at firms that already collected tips. Our design cannot separate an outright base-wage cut from a forgone raise; we interpret the clawback as the share of each dollar of induced tips offset by lower base pay, regardless of the mechanism.

¹⁶In the notation of Equation (3), the restriction is $\gamma_d = \gamma$. Pooling across doses requires this restriction since low-dose adopters serve as implicit controls for high-dose adopters. If there is heterogeneity in the dose coefficients, these dose comparisons can put negative weight on low-dose adopters (Balla-Elliott, 2026; Callaway et al., 2025; Graham and Powell, 2012). In Appendix E we re-estimate the response with a local least squares estimator that relaxes the constant-effect assumption and find similar results.

¹⁷Standard errors for the ratio are calculated from the joint covariance matrix of a stacked regression with both outcomes.

¹⁸Under the additional assumption that adoption affects pay only through the tips it generates, this ratio has an instrumental variable interpretation: the dose interacted with adoption timing instruments for induced tip income.

Combining the direct tax savings from Section 2 and the behavioral response from Section 3 with the clawback share estimated in this section, we find that the policy increased earnings net of federal income taxes by approximately 3.3% for the average tipped worker.¹⁹

5 Welfare: Marginal Value of Public Funds

The MVPF is the ratio of beneficiaries’ willingness to pay (WTP) for a policy to its net cost to the government (Hendren, 2016; Hendren and Sprung-Keyser, 2020). In our setting, the beneficiaries are workers and, through the clawback of base wages, firm owners; the net cost is the federal government’s forgone tax revenue less what it recoups on induced tips. NTOT’s MVPF lies between 1 (the *pure transfer* case) and 1.40 (the *tipping surplus* case, evaluated at the steepest step on the EITC phase-out schedule). The deduction delivers at least as much value per dollar of net fiscal cost as a lump-sum transfer. Appendix F details the construction of the MVPF.

5.1 The Welfare Value of the Induced Tips

The MVPF departs from 1 only if two conditions hold: NTOT must have changed behavior, and this behavioral change must have increased welfare. If NTOT did not change behavior, the deduction would be a lump-sum transfer to inframarginal recipients. Workers would value the \$531 average direct tax savings at face value, the government would bear that same amount as the policy’s mechanical cost, and the MVPF would be exactly 1. However, NTOT *did* change behavior; the average worker gained \$193 in induced tips (Section 3). This change in behavior may have increased welfare.

In the tipping surplus case, whatever motivates customers to tip (social norms, warm glow, the satisfaction of transferring money to people with higher marginal utility of income) fully compensates them for the money given, and the workers’ gains are new surplus. By revealed preference, then, customers would not be made worse off, and the tips would be new after-tax income to workers. The resulting MVPF would be 1.37. On the other hand, if tipping instead extracts money customers would rather have kept, each induced tip dollar costs the customer a dollar of welfare; net WTP equals the government’s net cost, and the MVPF is exactly 1.

The induced tips are new reported earnings, so they interact with the rest of the tax-and-transfer system. As reported earnings, they generate a fiscal externality: The government recoups \$ 16 per worker in payroll taxes, net of the present value of the Old-Age and Survivors Insurance liabilities those taxes create. The induced tips are also new earned income and enter the EITC schedules.²⁰

¹⁹Under 2025 schedules, a worker with mean earnings and tips (\$24,153 gross, \$6,215 in tips) owes \$840 in baseline federal income tax. Adding \$193 in tip income, subtracting \$54 in clawed-back base pay, and applying the NTOT deduction reduces the tax to \$213, a 3.3% increase in earnings net of federal income taxes.

²⁰The deduction itself reduces taxable income only: It alters neither AGI nor earned income. When behavior is unchanged, EITC, child tax credit, Supplemental Nutrition Assistance Program, and Affordable Care Act subsidy eligibility and payments do not adjust.

Each dollar raises the credit for workers in the phase-in range and lowers it for workers in the phase-out range. Because workers do not reoptimize to obtain this change, the envelope theorem does not net it out: It is a first-order transfer that moves WTP and net cost together. We can bound the EITC channel by applying the steepest segment in either direction. This moves the tipping surplus MVPF between 1.32 and 1.40.

5.2 Clawback and the Welfare Weights of Firm Owners

The clawback of base pay estimated in Section 4 means that the gains from the induced tips are split between workers and firm owners. The implications for the MVPF depend on the welfare weight placed on owners (Finkelstein and Hendren, 2020). With equal weight on the welfare of all private parties, the clawback relocates surplus rather than changing it. What workers lose in base pay, owners and the government gain between them.

Firms' response also creates no further first-order effects: By the envelope theorem, their reoptimization is second-order for their own welfare, and it consumes no real resources since firms turn tipping on with a few clicks in their POS settings. The response therefore matters only through its fiscal externality, the \$15 in payroll and income taxes the government no longer collects on the reduced base wages (Chetty, 2009). Workers retain the employee payroll-tax share and the forgone income tax, and owners keep the employer share, so WTP rises with net cost, and the MVPF is essentially unchanged at 1.36.²¹

If we instead weight owners' welfare at zero, so that the numerator counts only workers, their valuation of the induced tips is scaled by the retained share, and the MVPF declines to 1.25 (1.20 to 1.30 across the clawback's 95% confidence interval). Even under this workers-only accounting, the deduction delivers more value than a lump-sum transfer of the same net cost in the tipping surplus case. One caveat is that, to calculate the workers-only figure, we apply a clawback share estimated among the newly tipping firms to induced tips that arise at similar firms (Section 3).

6 Conclusion

NTOT raised the average tipped worker's earnings net of federal income taxes by approximately 3.3%, with the direct savings concentrated among the workers with the highest tip earnings. The policy also changed how firms pay: Firms newly enabled tipping, and of each induced tip dollar, workers kept 72 cents as a raise, while firms clawed back the rest through lower base pay. Because the induced tips are the policy's only behavioral response, the deduction delivers at least as much value per dollar of net fiscal cost as a lump-sum transfer. If the induced tips represent new surplus to workers at no welfare cost to customers (our tipping surplus case), the MVPF can be as high as 1.40.

²¹We ignore taxes on firm profits; including them would shift part of owners' gain to the government, raising the workers-only figure. Appendix F sizes this channel.

Our estimates of the clawback speak to the central question in debates about tipping: whether tips are a raise for workers or a wage subsidy to firms. In our setting, firms captured 28 percent of the induced tips through lower base wages. Both workers and firms were better off. Ours is the first estimate of the incidence of tips for firms that set wages, and it comes from variation generated by a single policy in one sample of small businesses that newly enabled tipping. A key task for future work is to estimate the incidence of tips from other sources of variation—and to better determine who benefits and what tipping means for workers and firms.

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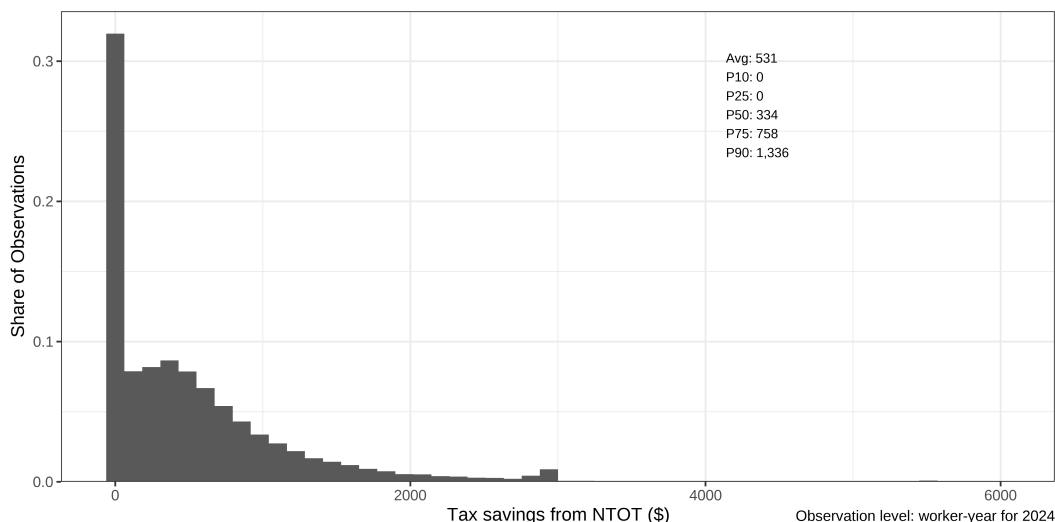
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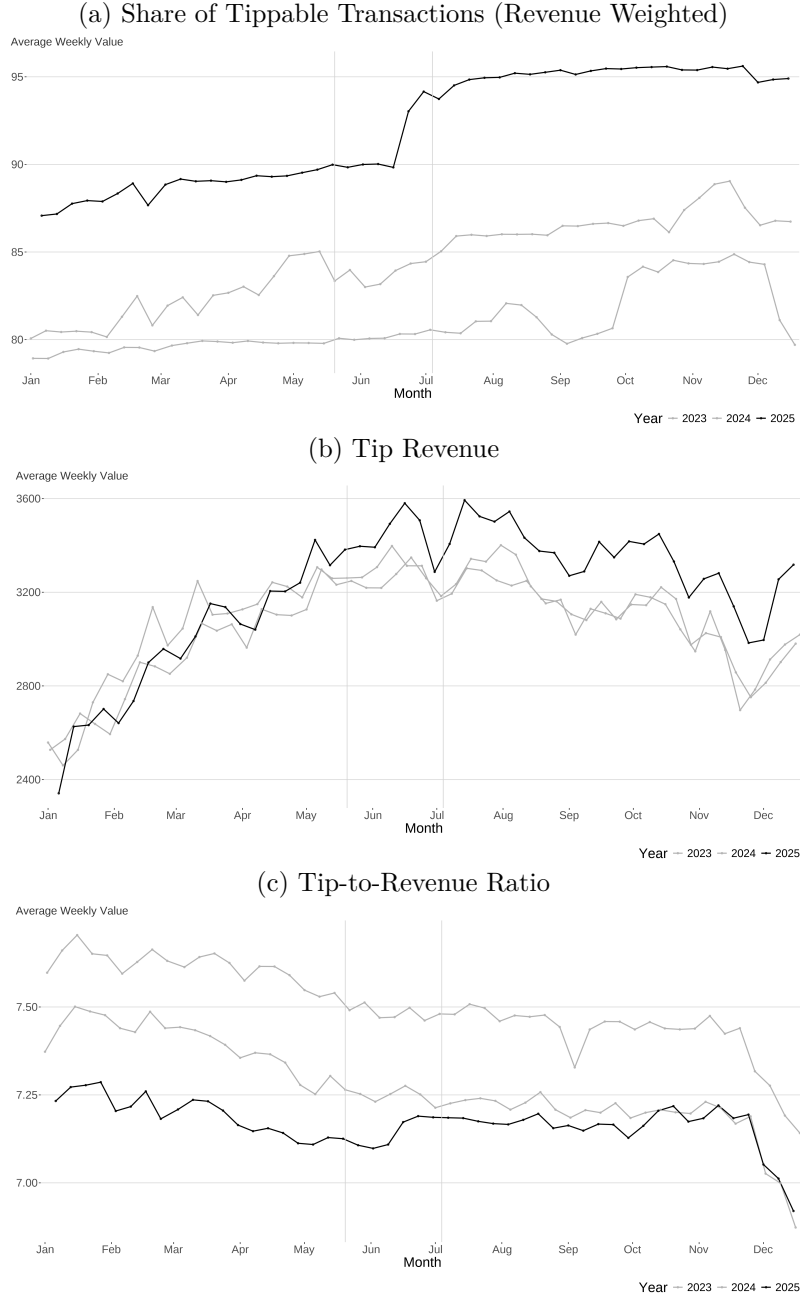
Figures

Figure 1: Distribution of Federal Income Tax Savings Under No Tax on Tips Provision



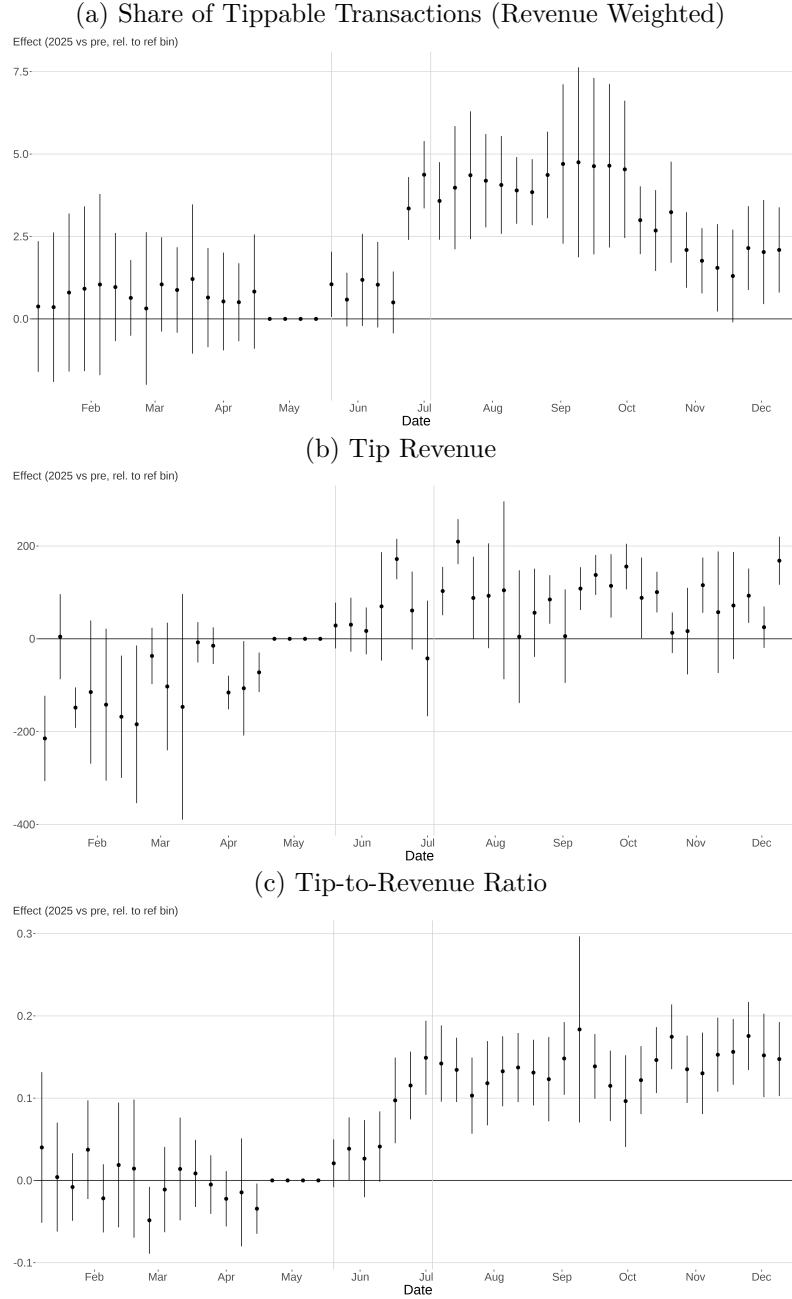
Notes: This figure presents the estimated distribution of federal income tax savings for tipped workers under the No Tax on Tips provision using 2024 earnings. Tax savings are calculated as the difference between federal income tax liability under the 2024 tax brackets and counterfactual liability with the NTOT deduction, assuming single filing status, standard deduction only, and no income sources beyond observed earnings. The distribution exhibits substantial heterogeneity: 26.4% of tipped workers receive zero tax savings, as their total earnings fall below the standard deduction threshold (\$14,600 in 2024), but the top quartile saves more than \$758 annually. Mean tax savings are \$531 per worker (median \$334). Values of \$3,000 and \$5,500 reflect workers who deduct the full \$25,000 tip income deduction at marginal tax rates of 12% and 22%, respectively. Among the 74% of workers with positive tax savings, mean benefits equal \$722 (median \$529).

Figure 2: Weekly Tipping Patterns By Calendar Year, 2023–2025



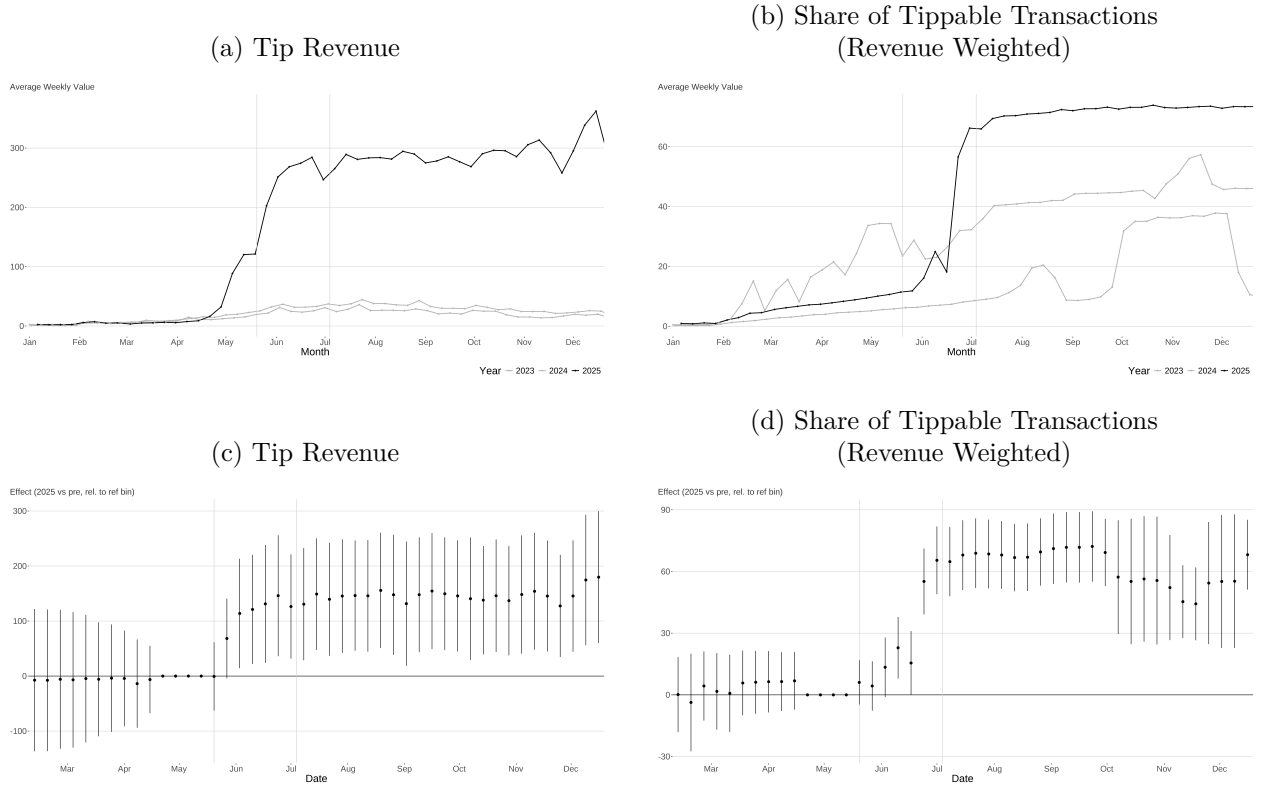
Notes: This figure presents weekly time series of tipping behavior for firms for 2023–2025, with 2025 observations displayed in black and those for prior years (2023–2024) in gray. The first vertical line marks May 20, 2025, when the Senate passed the standalone No Tax on Tips Act, and the second marks July 4, 2025, when the One Big Beautiful Bill Act containing the NTOT provision was signed into law. Panel (a) presents the revenue-weighted share of tippable card transactions. Panel (b) displays average weekly tip revenue per firm, and Panel (c) shows the ratio of tip revenue to total transaction amounts, both from all types of transactions. All measures represent unweighted simple averages across firms in each week. The last two weeks of each calendar year are excluded. The 2025 series begins diverging from historical patterns around the first policy date across the last two measures, and the share of tippable transactions has a particularly pronounced increase from 92% to 96% just before the provision was signed into law. These patterns provide initial evidence that the increased salience of tax-deductible tip income induced behavioral responses in tipping practices.

Figure 3: Estimates of Changes in Tipping Behavior Following No Tax on Tips Passage



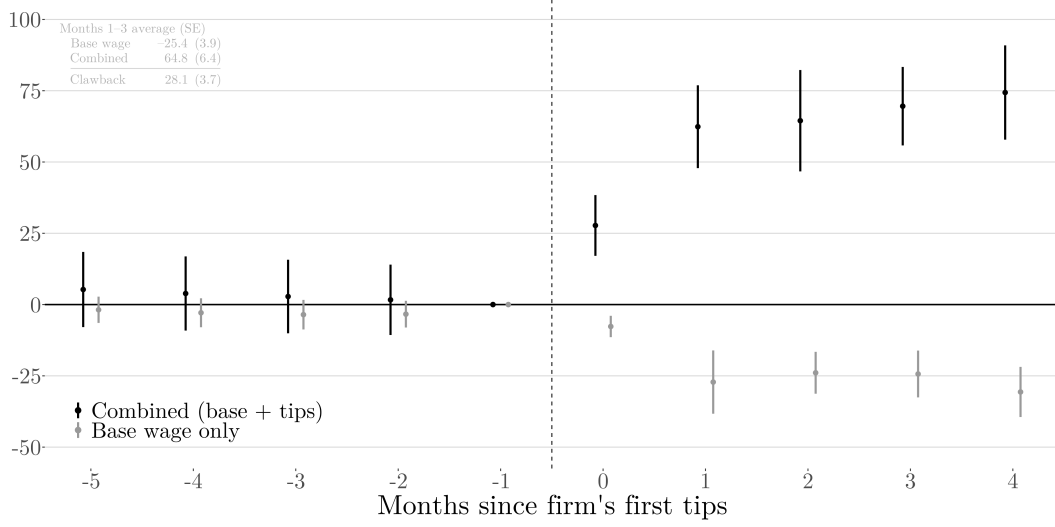
Notes: This figure presents event-study estimates of changes in tipping behavior following Senate passage of the No Tax on Tips (NTOT) Act. Under Assumption 1, each panel plots coefficients β_ℓ from Equation (2), which estimates the difference between 2025 outcomes and their calendar-time counterparts in 2023–2024, relative to the four weeks before May 20. The last two weeks of each calendar year are excluded. Vertical lines mark May 20 (Senate NTOT passage) and July 4 (One Big Beautiful Bill Act signing). Panel (a) presents the revenue-weighted share of tippable card transactions, showing an increase of 3–5 percentage points (pp). Panel (b) shows weekly tip revenue increased by approximately \$100 per firm following July 4. Panel (c) displays the tip-to-revenue ratio, which increased by roughly 0.15 pp. Tip revenue and the tip-to-revenue ratio are measured across all transactions; the share of tippable transactions is measured among card payments only. We show Driscoll and Kraay (1998) standard errors with a lag length of 8 weeks.

Figure 4: Tipping Behavior Changes in Firms with 0–5% Initial Tipping Share



Notes: This figure examines firm–years classified as having a revenue-weighted share of tippable card transactions of 0–5% in their first four weeks of each year. Firms can move between classifications across years on the basis of their initial four-week tipping intensity. For this analysis, we use the full firm roster rather than the top-decile analysis sample used in the overall sample figures. Panels (a) and (b) show weekly time series for 2023–2025, with 2025 in black and prior years in gray. Vertical lines mark May 20 (Senate passage of No Tax on Tips [NTOT]) and July 4 (signing of the One Big Beautiful Bill Act). Panels (c) and (d) present event-study estimates from Equation (2), with the four weeks preceding May 20 as the omitted baseline, excluding the first four weeks of each year (as these weeks define the subsample) and the last two weeks of each year (to avoid holiday-driven volatility). The share of tippable transactions increased by approximately 80 percentage points (pp) following NTOT’s passage, driving an increase of \$150 per week in tip revenue. These extensive-margin responses demonstrate that firms with minimal baseline tipping drove the aggregate effects. We show Driscoll and Kraay (1998) standard errors with a lag length of 8 weeks.

Figure 5: Effect of Tipping Adoption on Base-Wage and Combined Hourly Rates



Notes: This figure presents event-study estimates of the effect of a firm’s adopting tipping on the base-wage rate (gray) and the combined hourly rate (base wage plus tips per hour; black). A firm’s dose is its average tip income per hour over its first three full months of tipping (mean \$1.62, median \$0.75 among positive-dose adopters). Each series plots the dose-interaction coefficients β_k from Equation (5), measured relative to the month before adoption: β_k is the change in the outcome, in cents per hour, for each dollar per hour of tips the firm begins collecting. Zero-dose adopters receive a separate adoption effect and do not contribute to β_k . The sample contains 689 workers at 156 firms that adopted tipping between March and August 2025, with not-yet-treated firms as controls. Appendix E details sample construction, dose measurement, and the estimator. Post-treatment averages are taken over months 1–3; month 0 is the partial adoption month, and month 4 is displayed but excluded from the average. Over this window, β_k averages -25.4 cents (SE 3.9) for the base-wage rate and 64.8 cents (SE 6.4) for the combined rate. These averages imply a clawback of 28 cents per dollar of the induced tips (SE 3.7). Standard errors are clustered at the employer level.

Appendix

A Policy Background

The “No Tax on Tips” proposal originated as a campaign promise of both major candidates during the 2024 US presidential election. Google search interest for the term peaked in August 2024 during the campaign, but declined sharply thereafter. Public attention returned dramatically in May 2025, with search interest reaching its highest levels during the week of May 18th. Figure A1 presents Google Trends for search interest in the phrase “no tax on tips” from January 2023 to December 2025.

On May 20th and 22nd, 2025, the Senate’s “No Tax on Tips” Act and the House’s “One Big Beautiful Bill” Act (OBBBA) cleared their respective chambers. Prediction markets reflected the growing likelihood of enactment: Polymarket odds for “Trump ends taxes on tips in 2025?” rose from 78% on May 12th to 95.5% by May 21st,²² while odds for “End Tax on Tips” being included in the next reconciliation bill increased from 72% on May 14th to 92% by May 21st.²³ The OBBBA version passed the Senate on July 3rd and was signed into law on July 4th, 2025. The “No Tax on Tips” (NTOT) provision allows workers to deduct up to \$25,000 of qualified tips from federal income taxation starting retroactively in 2025. The provision is scheduled to expire after 2028. The deduction applies only to federal income taxes, not payroll taxes, and phases out at higher income levels.

All existing estimates rely on imputed or self-reported tip data, which may not accurately reflect the actual tip income of affected workers. These measurement challenges operate through multiple channels. Self-reported survey data like the Survey of Income and Program Participation (SIPP) rely on individual recollection of tip income, which may be subject to recall error. Tax return data from W-2 forms capture only reported tips, missing unreported cash tip income. Imputation methodologies rely on observable correlates of tip income that may not fully capture heterogeneity in tip earnings across firms, occupations, and workers within the same industry classification.

The absence of directly observable tip earnings linked to administrative tax records represents a critical gap in the evidence base for evaluating NTOT. This paper addresses this limitation by leveraging payroll and point-of-sale records that capture actual tip transactions. This approach enables the first empirical estimates of both the policy’s direct distributional effects and behavioral responses to tax changes.

²²Source: <https://polymarket.com/event/trump-ends-taxes-on-tips-in-2025>

²³Source: <https://polymarket.com/event/what-will-be-in-the-next-reconciliation-bill>

B Data, Variable Definitions, and Representativeness

B.1 Overview of Data Sources

We use administrative data from a large business technology platform. The data cover firms primarily in the food and beverage and retail sectors through their payment processing and payroll records. Our analysis exploits linked point-of-sale and payroll records covering firms that appear in both data sources. The sample covers all 50 states from 2023 through 2025. Our dataset offers distinctive advantages for studying the labor market and behavioral impacts of tax policy changes affecting tipped workers.

First, the linked structure of point-of-sale and payroll records provides comprehensive measurement of both firm revenue flows and worker earnings. This administrative linkage allows us to observe tip income as it appears in formal payroll records while also enabling validation of reporting consistency between point-of-sale and payroll records for firms appearing in both data sources.²⁴ Second, the high-frequency nature of our data allows us to observe tipping behavior in real time around policy passage, enabling estimation of immediate behavioral responses to the legislation.

B.2 Data Structure and Observable Variables

The de-identified data include consistent firm identifiers that allow us to link the same business across both payroll and point-of-sale products.

Transaction-level point-of-sale records include itemized receipts with quantities, prices, payment methods, subtotals, and tip amounts. We also observe whether tipping was enabled for a particular transaction. These granular records capture both intensive margin changes (tip amounts per transaction) and extensive margin responses (adoption of tipping technology).

Payroll records include individual worker paystubs with detailed compensation breakdowns. Each paystub contains distinct fields for base wage rates, hours worked, overtime premiums, and tip distributions. The detailed nature of the payroll records allows us to aggregate paystubs to the monthly level and construct annual earnings profiles for workers while preserving within-year variation in employment and compensation patterns.

For the direct effects analysis, we construct annual worker earnings by aggregating monthly payroll records. This allows us to analyze the distribution of tax savings under the NTOT provision. We calculate each worker’s tip earnings share and apply 2024 federal tax schedules to estimate individual tax savings under the NTOT provision. We assume single filing status, that filers take only

²⁴While our main analysis does not restrict the sample based on tip reporting validation, this capability enables quantification of tip under-reporting rates in payroll records. It also provides a framework for future research examining whether the NTOT provision induces workers to shift a larger share of tip earnings into formal payroll channels. Such reporting responses would partially offset the fiscal costs of the provision by expanding the tax base for tip income, albeit at lower marginal rates.

the standard deduction, and that they have no other income sources. These assumptions provide a baseline for quantifying the provision’s direct fiscal effects.

B.3 Representativeness

The point-of-sale and payroll data primarily cover small and medium-sized firms in the food and retail sectors. These firms rely on external technology providers for payment processing and payroll, rather than developing proprietary systems. This adoption pattern is characteristic of small and medium-sized enterprises. Like other administrative data sources (e.g., the large payroll provider used in Grigsby et al. (2021)), our data do not capture the largest firms in the economy, which typically develop proprietary software systems rather than using external software providers.

B.4 Comparison to Existing Data Sources

Tips are notoriously difficult to measure since they are typically self-reported. Existing data sources have significant limitations. Survey data from the SIPP and CPS (which combines tips with overtime and commissions) rely on self-reports that may misstate tip income. Administrative tax data from IRS Forms W-2 capture only reported tips, missing substantial unreported amounts: Basker et al. (2024) estimate that \$8 billion in tips paid annually at full-service, single-location restaurants were not captured in tax data from 2005 to 2018. Private payroll providers like ADP observe tip distributions but lack the transaction-level detail needed to measure tipping behavior.

The closest work in this respect is found in the recent studies that have used internal records of tips from Uber, which are directly recorded through the payments from riders to drivers (Chandar et al., 2024a,b). Our paper maintains similar advantages associated with using business records, but provides new estimates for “brick-and-mortar” firms.

C Direct Impacts

The direct impact analysis in Section 2 examines the distribution of tax savings for tipped workers using 2024 payroll data (before the policy took effect) to isolate direct fiscal effects from behavioral responses. This analysis uses worker-level annual earnings as the unit of observation.

The direct effects sample comprises tipped workers who have complete 2024 earnings records and meet minimum work attachment thresholds. To construct annual earnings estimates for all workers rather than only those observed for 12 months, we employ a donor-matching procedure that pairs partial-year observations with comparable full-year workers based on geographic location, industry classification, and observable worker and firm characteristics. Detailed sample construction procedures and validation checks appear in Appendix C.1.

C.1 Sample construction

The NTOT provision allows for qualified tips to be deducted from income for federal tax purposes. Since the law was enacted mid-year, we might be worried about reclassification of non-tipped workers into tipped roles or workers shifting their income from base wages to tips. To estimate the direct impacts of the law without having to account for these behavioral responses, this analysis uses data from 2024, before the policy applied. To do so, we use payroll records, where individual paystubs are reported at the firm level for each employee. We aggregate these paystubs to the monthly level based on the date of payment.

On each paystub, we observe gross pay for the pay period as well as its constituent components. We observe base wages and hours; overtime and double-time wages and hours; and any additional compensation such as bonuses or tip earnings.

To construct a representative sample of annual earnings for tipped workers, we impose several restrictions. We begin by subsetting to workers who have any tip earnings in 2024. We then restrict to worker-firm-months with at least 5 hours worked and require each worker to appear for at least 2 months. A unique feature of our data is the ability to track individuals across employment relationships, allowing us to include months without tip earnings for workers who received tips at other firms during the year.

Tipped workers appear in our data for varying durations in 2024. We define two baseline samples: workers observed for all 12 months and workers observed for fewer than 12 months. To estimate yearly earnings for all workers rather than only the 12-month subset, we construct a combined sample using donor-matching. For each worker-firm-month in the partial-year sample, we construct a donor pool of 12-month observations who share the same ZIP3, industry, and sub-industry. We then calculate the Euclidean distance on standardized worker- and firm-level monthly variables.²⁵ We keep donor-recipient matches where the monthly distance is less than 3 in all months to ensure match quality and stability. Among qualifying matches, we keep the 10 donors with the lowest average Euclidean distance across months.

Figure A2 shows the earnings distributions for the three samples. The combined sample places greater weight on lower-earning workers from the donor pool, reflecting that partial-year workers tend to have lower annual earnings than those observed for the full year. This weighting ensures our final sample appropriately represents the range of tipped workers in our data, not just those with stable year-long employment.

C.2 Characteristics and distribution of savings

This appendix collects the worker- and firm-level distributions summarized in Section 2.

²⁵The variables are monthly earnings, monthly hours, base wage, tip earnings for workers; and total revenue, tip-based revenue, and firm size for the firm. For any firm-level variables we do not observe, we remove them from the calculation and use the remaining variables.

Table A1 reports summary statistics for workers and firms in the combined sample. Panel A describes job characteristics across worker-firm-months in 2024: average monthly earnings are \$1,970, composed of 90 hours at roughly \$22 per hour. Panel B describes the firms. Table A2 reports the firm size distribution: 82.4% of firms have fewer than 15 workers and 64% have fewer than 10, so that approximately three-quarters of workers are at firms with fewer than 20 employees.

Figure A3 presents the distribution of annual earnings and tip composition. Figure A3a shows that yearly income is right-skewed, with a median of \$20,950 and a 90th percentile of \$42,135, placing the sample at the lower end of the earnings distribution. Figure A3b shows the share of annual earnings coming from tips: the mean is 26%, and the distribution has a long right tail, with the top 10% of workers earning more than 46% of their income from tips and the bottom 10% earning 7% or less.

Figure A4 expresses each worker’s tax savings as a share of annual earnings. Average savings are 1.7% of earnings, and the mass at zero corresponds to workers below the standard deduction threshold. Figure A5 plots the concentration curve for tax savings, the cumulative share of aggregate savings against the cumulative share of workers ranked by 2024 earnings: the top 20% of workers capture 51% of aggregate savings while the bottom 50% receive 11.3%.

C.3 Robustness in the Validated-Tip Sample

The linked structure of our data enables direct validation of tip reporting consistency between the point-of-sale and payroll records. This appendix presents distributional analysis using a validated subsample restricted to workers at firms where we observe both transaction-level revenue flows and worker-level payroll records. This linked sample addresses measurement concerns about tip reporting accuracy and provides evidence on the reliability of tip income measures used in the main analysis.

C.3.1 Sample Construction and Validation Criteria

The validated sample imposes two firm-month-level quality criteria beyond the baseline restrictions in Appendix C.1. Each criterion serves a distinct validation purpose:

Earnings-to-Revenue Ratio. We verify that monthly payroll expenses remain within a plausible range relative to monthly revenue. Specifically, we exclude firm-months where total payroll earnings exceed 120% of firm revenue, which suggests data quality issues or measurement error. This threshold accommodates service industries where labor costs represent substantial revenue shares while identifying implausible observations.

Tip Validation. For each firm-month, we assess correspondence between point-of-sale tip collection and payroll tip distribution. We classify firm-months as validated when: (i) both tip measures equal zero, (ii) tip revenue is zero while payroll tips are positive (acceptable given potential timing differences), or (iii) both measures are positive and the ratio of tip revenue to payroll tip earnings

falls between 0.8 and 1.2. This range accommodates normal timing differences in tip collection and distribution while flagging substantial discrepancies.

Workers appear in the validated sample if they work at qualifying firm-months, have non-zero tip earnings during the year, and appear for all 12 months of 2024 across firms. This stringent selection yields a smaller sample in which under-reporting of tip earnings is not a concern. The validated sample therefore provides a stringent check on the main analysis.

C.3.2 Sample Characteristics and Selection Patterns

Table A4 presents worker and firm characteristics for the validated sample. The validated sample exhibits systematically higher earnings and larger firm sizes than the main sample, reflecting selection into firms that appear in both the point-of-sale and payroll data. Job-months in the sample average \$2,575 in monthly earnings (31% higher than the main sample’s \$1,970 per job), with 106 hours per month compared to 90 hours in the main sample. Hourly wage rates average \$23.94 versus \$21.77 in the main sample. These differences emerge mechanically from the sample construction, rather than reflecting fundamental differences in tip-earning populations.

Firms in the validated sample demonstrate substantially higher scale and formality. Average firm size is 12.53 workers (vs. 9.88 in the main sample), with monthly firm revenue averaging \$76,453 (vs. \$52,391). However, per-unit economics are similar across samples: revenue per hour (\$82.75 vs. \$84.08), mean firm hourly wage (\$23.52 vs. \$23.33), and full-time share (35.0% vs. 32.4%) are nearly identical (Tables A1 and A4). The validated sample thus differs from the main sample primarily in scale, supporting the comparability of the validated estimates despite the selection on firm size.

The selection of larger, higher-revenue firms into both data sources reflects the costs and scale economies of adopting external business software. Critically, the validation criteria further select firms that demonstrate consistency between transaction-level tip collection and payroll-reported tip distributions. This selection distinguishes the validated sample from firms where tip reporting may be incomplete or where alternative tip distribution channels operate outside formal payroll systems. The validated sample thus captures firms with both the scale to adopt external business software and the practice of formally reporting tip income through payroll.

C.3.3 Distribution of Tax Savings

Figure A6 presents the distribution of estimated tax savings from the NTOT provision in the validated sample. Average tax savings are \$856 per worker, compared to \$531 in the main sample. This 61% difference reflects the validated sample’s higher earnings levels rather than differences in tip intensity.

The validated sample exhibits different patterns of benefit receipt. Only 18.5% of workers receive zero tax savings because earnings fall below the standard deduction threshold, compared to 26.4% in the main sample. This difference reflects selection of higher-earning firms into appearing in both

the point-of-sale and payroll data. Workers at smaller, lower-revenue firms face higher probabilities of earning below tax filing thresholds.

Among workers with positive savings (81.5% of the validated sample versus 74% in the main sample), average savings are \$1,051 (median: \$788). The distribution exhibits substantial dispersion: the 25th percentile of positive savings is \$349, while the 75th percentile reaches \$1,498. The 90th percentile is \$2,406. Clear spikes appear at \$3,000 and \$5,500, corresponding to workers deducting the full \$25,000 tip income deduction at the 12% and 22% marginal tax rates.

The top 20% of workers in the validated sample (ranked by total earnings) capture 42% of aggregate tax savings, compared to 51% in the main sample. The lower concentration follows mechanically from the validated sample’s higher earnings floor: with fewer workers below the standard deduction threshold, benefits spread across more of the distribution.

C.3.4 Validation Implications

The validation exercise yields two key findings:

Data Quality. The ability to link transaction-level tip collection to payroll-reported tip distributions provides direct evidence of reporting consistency. Within the validated sample, point-of-sale tip revenue and payroll-reported tip distributions correspond within a 20% tolerance, providing direct evidence that the tip earnings used in our main analysis are consistent with tips actually collected. This validation addresses a fundamental limitation of prior research relying on self-reported survey data or tax returns with partial tip reporting.

Distributional Heterogeneity. The validated sample’s concentration metrics (top 20% capturing 42%, 18.5% receiving zero benefit, \$1,051 average savings among beneficiaries) all shift in the direction predicted by its higher earnings distribution. With more workers above the standard deduction threshold, benefits are less concentrated and average tax savings rise. These differences reflect the composition of firms appearing in both data sources rather than substantive variation in how the policy treats tipped workers. The concentration pattern itself, that higher earners capture disproportionate savings, holds in both samples.

The validated sample’s departures from the main sample are explained by its higher earnings floor rather than by any substantive difference in the policy’s distributional effects. The validation exercise demonstrates that our point-of-sale and payroll records provide consistent measures of tip income, supporting the interpretation of our main findings as reflecting the policy’s effects on tipped workers more broadly and improving upon prior data sources.

C.4 Comparison to Existing Estimates and Distributional Effects

Existing estimates of NTOT’s impact indicate highly concentrated benefits among higher-income tipped workers. Only 2% of all households benefit from the legislation, and among tipped workers specifically, just 60% of households see tax reductions (Gleckman, 2024). This limited reach reflects

the fact that 26.4% of tipped workers in our sample already pay no federal income tax, making the deduction worthless for these lower-income workers.

The Council of Economic Advisers estimates an average tax cut of \$1,675 per tipped worker, calculated by dividing the Joint Committee on Taxation’s revenue estimate (\$10.1 billion annually) by IRS tabulations of tip recipients (6 million workers) (Council of Economic Advisers, 2025). The Urban-Brookings Tax Policy Center projects similar savings of \$1,800 for tipped workers who currently owe federal taxes (Gleckman, 2024).

The Budget Lab at Yale (TBL) released a September 2024 report that provides the most directly comparable estimates to our analysis (The Budget Lab, 2024). Their methodology represents the most sophisticated existing approach to addressing data limitations. They use random forest algorithms to impute tip income in IRS tax microdata from 2017, starting with SIPP data to estimate relationships between tips, industry, wages, and demographics. They then benchmark total tip earnings to IRS W-2 data from 2017. They find only 4 million workers in tipped occupations (2.5% of employment), with median weekly wages of \$538 versus \$1,000 for non-tipped workers.

TBL estimates that families who benefit would receive an average tax cut of approximately \$1,700. However, their distributional analysis reveals extreme inequality in benefits: bottom-quintile tipped workers receive only \$200–450 on average, while top-quintile workers average \$5,768 in tax savings.

Our approach differs from TBL’s in several ways. First, we observe actual tip earnings rather than imputing them based on relationships between survey data and demographic information. Second, TBL’s key finding uses 2026 tax brackets under Tax Cuts and Jobs Act (TCJA) expiration, while our main estimates use 2024 brackets with TCJA provisions (which were extended in the OBBBA). Third, our main estimates include the \$25,000 tip deduction cap and income phaseout added in the OBBBA legislation, which TBL did not incorporate as their report predates passage.

To facilitate direct comparison, we recalculate our estimates using 2026 brackets assuming TCJA expiration, then convert these to 2024 values by deflating using the C-CPI-U (reversing the IRS’s methodology for calculating annual bracket adjustments). Under our methodology, without the deduction cap and income phaseout, and conditioning on workers who benefit, the average worker saves \$847 in taxes (median \$553). When we classify workers by ACS household money-income quintiles, the bottom-quintile tax cut for those with any benefits is \$517, 2.6 times the TBL estimate of \$200.²⁶

TBL’s Figure 6 presents average tax cuts by income quintiles for 2025 under TCJA tax brackets for those who benefit. Since both 2024 and 2025 use TCJA brackets, our estimates are directly comparable to these figures. To match TBL’s specification, we calculate tax savings without the income phaseout and tip deduction cap. Using ACS household money-income quintiles, our estimated tax savings for the bottom quintile are approximately \$475, compared to TBL’s \$220. For

²⁶Using the Tax Policy Center’s “expanded cash income” percentile breaks, average tax savings for those in the bottom quintile increase to \$530.

middle-quintile earners, we calculate tax savings of \$2,762, compared to their \$1,670. For highest-quintile earners, we calculate \$6,851 versus their \$5,820. Despite our sample of over 130,000 workers, we observe only 300 individuals in the top two income quintiles. Including the cap and phaseout reduces tax savings by 60% for top-quintile earners and by 17% for middle-quintile earners.

Several factors may explain the systematic differences between our estimates and TBL’s. First, our sample is heavily concentrated at the lower end of the income distribution. Second, TBL uses IRS microdata that better accounts for household structure and defines income more comprehensively, while we observe only individual workers and assume no income beyond our data. Third, our observed tip earnings may be more accurate than SIPP-based imputations, potentially explaining our consistently higher estimated benefits, particularly for low-income workers. Finally, our sample comes from firms that use external payroll services and are concentrated in food service, capturing a substantial portion of tipped employment.

D Behavioral Response

This appendix presents the subsample and robustness exhibits supporting the behavioral analysis of Section 3.

D.1 Sample construction

The behavioral response analysis in Section 3 examines changes in firm tipping practices following the policy’s passage, aggregating transaction-level data to weekly firm-level measures of total tip revenue, tip-to-revenue ratios, and the revenue-weighted share of tippable transactions. This panel covers 2023 to 2025.

The behavioral sample restricts to firms with continuous monthly activity throughout 2023–2024 and sufficient pre-period transaction volume to ensure statistical power for detecting policy-induced changes in tipping behavior. All selection criteria use 2023–2024 activity only; the panel itself runs through December 2025. Specific eligibility criteria and the rationale for transaction-volume thresholds appear below.

Our data allow us to observe every component of an itemized receipt, from prices and quantities sold to payment method, subtotals, and tip amounts. Importantly, we can measure whether tipping was enabled for transactions made by credit/debit cards and bank transfers. Cash transactions only record whether a tip was received by the store, and as such we exclude them, though they make up a small share of revenue (9% on average across firms).

Sample construction proceeds in two stages. First, we define a firm roster using 2023–2024 activity only, keeping firms that (i) have at least one card transaction in each of the 24 pre-period months, (ii) average at least 25 transactions per day over 2023–2024, and (iii) average at least \$250 per day in revenue over 2023–2024. Second, we aggregate all transactions for these firms into a weekly panel

spanning January 2023 through December 2025. Within each firm’s observed range, missing weeks are filled with zero activity to ensure continuous within-firm time series. For our main analysis sample, we further restrict the roster to firms in the top decile of average weekly card-transaction count during 2023–2024, which corresponds to roughly 1,000 or more weekly transactions. The subsample analyses (food and beverage, retail, “near zero”, and “near one”) drop this top-decile restriction and use the full roster, which by construction has at least 175 weekly transactions on average (25 per day times 7 days in a week). Because all selection criteria are computed on 2023–2024 data, sample composition is not conditioned on firm survival.

This overall sample consists of approximately 5,700 firms that have an average weekly transaction count of 2,361 as seen in the Overall column of Panel A of Table A3. These restrictions reduce the sample of firms to those with an average weekly gross transaction volume (revenue plus sales tax and tips) of \$57,900, on the upper end of the broader firm population in the underlying data. The average transaction has a bill size (total sale amount including subtotal, tax, and any tips) of \$20.27. Among tipped transactions, the average tip is \$2.93, or roughly 17% of the subtotal amount.

Table A3 reports transaction-level descriptive statistics for the behavioral samples, split by industry and by baseline tipping intensity. The “Overall” column restricts to the top decile of weekly card-transaction count, while the other columns use the full firm roster. Food and beverage firms (the “Food & Bev” column) average roughly \$13,560 in weekly gross transaction volume with \$1,066 from tips: lower volume than the Overall column but broadly similar bill sizes and tip percentages. The “Near Zero” and “Near One” columns classify firm-years by whether the revenue-weighted share of tippable transactions in the first four weeks of the year falls in 0–5% or 95–100%, respectively, and the two groups differ sharply in tip revenue, the share of tippable transactions, and average bill size.

D.2 Heterogeneity by industry and baseline tipping

The aggregate response documented in Section 3.3 masks heterogeneity across industries and across firms’ baseline tipping intensity. All subsample analyses use the full firm roster rather than the top-decile analysis sample.

Figure A10 replicates the time series and calendar-time event study for food and beverage firms. The share of tippable transactions rises sharply around the NTOT date and the tip-to-revenue ratio rises at the same time. Under Assumption 1 (CTCT), the share of tippable transactions increases by 2 percentage points and the share of revenue coming from tips increases by 0.10 percentage points, directionally consistent with the overall sample.

Figure A11 presents the same analysis for retail firms. Under Assumption 1 (CTCT), the share of tippable transactions increases by 15 percentage points and the tip-to-revenue ratio increases by 0.02 percentage points; the tip-to-revenue series is noisier but suggestive of a post-NTOT increase.

Figure A12 examines firm-years that begin the year with 95–100% of revenue-weighted transactions

already tippable. Tip revenue shows no significant change (Figure A12a). Although these firm-years started the pre-period at a 98.6% share of tippable transactions, Figure A12b shows that share remained elevated throughout 2025 while in earlier years it declined steadily over the year. Under Assumption 1 (CTCT), we estimate no statistically significant change in the revenue-weighted share of tippable transactions (Figure A12d), consistent with these firm-years already operating near full tipping saturation. The near-zero firm-years, which drive the aggregate response, are discussed in Section 3.3 and shown in Figure 4.

D.3 Placebo and event-study robustness

Two exercises support the interpretation of the main event study. The first checks that the tip response does not reflect changes in overall sales; the second checks that the calendar-time comparison is not picking up spurious seasonality.

Figure A8 plots event study estimates for revenue and gross transaction volume (revenue plus sales tax and tips), the sales-side counterparts to the tipping outcomes. Under Assumption 1 (CTCT), neither measure shows a statistically significant change following NTOT passage, though the confidence intervals admit moderate movements in either direction. Because sales are flat while tip revenue rises, the increase in tip revenue reflects greater tipping availability and higher tip-to-revenue ratios rather than increased customer traffic or larger purchases.

Figure A9 re-estimates Equation (2) treating May 20th, 2024, as a placebo “policy” date and using 2023 as the control year. No panel shows a systematic positive shift following the placebo date. The estimates display some noise and imperfect pre-trends, particularly for the revenue-weighted share of tippable transactions (Figure A9c), which shows negative coefficients early in the calendar year, but there is no evidence of a positive shift in the weeks after the placebo date. These null results contrast with the significant 2025 effects and support Assumption 1 (CTCT).

E Estimating the Clawback

This appendix describes the sample, the dose, and the estimation behind Section 4. Under common trends and a constant dose slope, the coefficient in the body is the clawback per dollar of induced tips. We test and relax both assumptions in turn; the results are similar, because both assumptions are supported by the data.

Figure A13 presents the unconditional adoption event study, which omits the dose interaction and plots only the adoption-month dummies δ_k from Equation (5). The base-wage rate is near zero with tight confidence intervals before adoption; afterward, the point estimates decline but the standard errors widen sharply, from roughly 0.05 pre-adoption to 0.22–0.48 post-adoption. The combined rate displays wide confidence intervals throughout; they narrow briefly at adoption before returning to pre-period levels. This pattern reflects heterogeneity in the treatment effect across firms with different tipping intensities, not a violation of common trends: the unconditional specification

averages over the dose distribution, and the dose-response design addresses this heterogeneity directly.

E.1 Sample and Dose

The sample includes firms with at least 24 months of payroll records over 2023–2025, no tips collected at any point in 2024, and a first month of tip collection in 2025. The adoption month is the first month with any tips collected. Firms adopting between March and August 2025 form the treated cohorts. We use not-yet-treated firms as controls; September and October adopters enter as controls only. On average, firms are observed with both payroll and point-of-sale records for 17.0 months before adopting tipping.

From this sample of tipping-adoption firms, we construct a worker-level panel of their employees. Each worker must be observed in every month from relative months -3 to 2 around the firm’s adoption (6 months total), and months outside this window enter the estimation when observed. Later event months are identified from fewer cohorts: relative month 2 excludes the August adoption cohort, whose month 2 falls in October when no not-yet-treated controls remain, and relative month 3 is identified from the March–June adoption cohorts only (103 of 156 firms). The sample contains 689 workers at 156 firms.

An adopter’s dose is the average tip income per hour over its first three full months of tipping.²⁷

Among positive-dose adopters, the mean dose is \$1.62 per hour and the median is \$0.75; worker-weighted moments are nearly identical. The dose distribution is highly skewed because 45% of adopting firms enable tipping in month 0 but collect essentially no tips over months 1–3, producing a dose of zero; these firms employ 48% of sample workers.²⁸ In our main specification, zero-dose adopters enter as controls but do not contribute to the dose effects; excluding zero-dose adopters entirely leaves the slope essentially unchanged. Allowing zero-dose adopters to also identify the slope, instead of assigning them a separate effect, yields 23 cents (95% CI [13, 33]).

The clawback sample and the behavioral sample cover the same economic margin (firms that begin collecting tips because of NTOT) but are constructed with different restrictions. The behavioral sample selects on POS transaction volume while the clawback sample selects on payroll coverage and pre-period tip absence. Both capture firms that begin collecting tips because of NTOT. The \$193 in annual induced tips is estimated across all behavioral-sample firms, including those that were already tipping before NTOT. Because the response is concentrated among newly tipping

²⁷We take the larger of the tips reported in its payment records and its payroll records. In a firm’s first months of tipping, the two records can come online at different times: a firm may begin collecting tips at the point of sale before routing them through the payroll product, or distribute early tips through payroll while point-of-sale tipping is still being configured. Taking the larger of the two records makes the dose robust to this onboarding timing, and the adoption month is the first month with tips in either record, so a lag in one system can neither misdate adoption nor understate the dose.

²⁸Over the full sample of adopters, the mean dose is \$0.89 per hour and the median is \$0.04. Recall that zero-dose adopters remain in the sample as controls: they contribute to the common trends comparison as not-yet-treated firms but receive a separate adoption dummy and do not contribute to the dose coefficient of interest.

firms, per-adopter induced tips exceed this average. Applying the clawback share to the population average therefore understates the dollar magnitude of the clawback.

E.2 Estimation and Robustness

The stacked design is built from “slices”: for each adoption cohort, we pair its workers with workers at firms not yet treated during its event window (Deshpande and Li, 2019).

Within each slice, we estimate Equation (5) at the worker level. The dose slope is fully interacted with cohort and relative time, so each cohort-by-event-month cell has its own dose coefficient. We aggregate the cell-level coefficients across adoption cohorts in proportion to the number of positive-dose treated firms. Standard errors are clustered on the employer.

Under common trends and the assumption that the response is constant in the dose, the coefficients recover the effect of each dollar of dose. We test and relax each assumption in turn.

E.2.1 Relaxing Common Trends

The dose does not predict pre-adoption changes in either outcome (base-wage rate: -2.8 cents, $p = 0.28$; combined rate: -4.0 cents, $p = 0.22$). One may still worry that firms with different doses are on different trends. As a relaxation, we interact the dose with the time fixed effects, allowing outcomes to trend differentially by dose. The implied clawback is 26 cents (95% CI [18, 33]), compared to our baseline of 28 cents in the body.

The choice of control pool is a further check on common trends, because it changes which firms identify the counterfactual time effects. Our baseline control pool is every not-yet-treated firm: March–August adopters contribute control months before their own adoption, and September and October adopters, contribute their not-yet-treated months, which identify the time trends for August and September. Extending the pool to firms adopting in November and December 2025 identifies the time effects through November, restoring month-3 identification for the July cohort and months 2 and 3 for August; the clawback under the extended pool is still 28 cents (95% CI [18, 38]). We exclude these late adopters from the baseline because policy salience declines after August, making their adoption motives less clearly tied to NTOT.

E.2.2 Relaxing Constant Effects: Local Least Squares

If the dose coefficient varies with the dose, the OLS specification weights dose levels unevenly and can weight some negatively. The regressor that carries the dose variation is the interaction $X_{it} = D_{j(i)} \times \text{adopted}_{it}$: over adoption, X_{it} rises by the dose for adopters and stays at zero for not-yet-treated controls. Balla-Elliott (2026) shows that the pooled slope is a weighted average of the dose-specific responses with implied weights

$$\omega_i \propto \Delta X_i (\Delta X_i - \mathbb{E}[\Delta X_i]) : \quad (6)$$

Adopters with below-mean doses can receive negative weight, with low-dose adopters implicitly serving as controls for high-dose adopters. The same problem arises in difference-in-differences designs with a continuous treatment (Callaway et al., 2025) and in the correlated random coefficients model of Graham and Powell (2012).

To relax this assumption, we use the local least-squares (LLS) estimator of Balla-Elliott (2026). For any dose $d \neq 0$, the response at d is the slope of a feasible local regression that uses only adopters at dose d :

$$\gamma_d = \frac{\text{cov}(\Delta Y_i, \Delta X_i \mid \Delta X_i \in \{0, d\})}{\text{var}(\Delta X_i \mid \Delta X_i \in \{0, d\})}. \quad (7)$$

Iterating over the dose distribution and averaging recovers an equally weighted average response. The not-yet-treated controls supply the $\Delta X_i = 0$ group; each local regression then makes exactly the comparison in Equation (3) of the body. In practice, each local regression uses adopters with doses within a bandwidth of 0.40 of d , averaging over 58 treated firms against 64 control firms, with confidence intervals from a normal approximation using firm-level bootstrap standard errors.

The LLS clawback is 39 cents (95% CI [13, 66] cents), consistent with the pooled estimate of 28 cents. Figure A14 plots the dose-specific responses for the base-wage rate and the combined rate. The LLS point estimate sits above the pooled one because the per-dollar clawback is smaller among the highest-dose firms, which receive more weight in OLS; the dose-specific estimates display this convexity. However, the differences are not statistically significant.

F Construction of the MVPF

This appendix details the construction of the marginal value of public funds (MVPF) in Section 5: the formulas, their components, the clawback adjustment, and the channels we do not model.

F.1 MVPF setup

For worker i , let M_i denote direct tax savings, T_i tip earnings in 2024, and $dT_i = \rho T_i$ induced tips, where ρ is the proportional tip response estimated in Section 3. Let g_i be the payroll taxes recouped by the government on induced tips and r the earned income tax credit (EITC) slope the worker faces. Employees and employers each owe Federal Insurance Contributions Act (FICA) taxes at $\tau_e = 7.65\%$, so the combined rate is $\tau_f = 2\tau_e = 15.3\%$. Applying ρ to the incumbent tip earnings reconstructs total induced tips, including those at newly tipping firms, because ρ is the aggregate response computed over all firms, so new firms' tips load onto the incumbent base; adding a separate term for newly tipping firms would double count.

The MVPF is a ratio of sums, never an average of worker-level ratios. The denominator is the

government's net cost,

$$D(r) = \sum_i [M_i - g_i + r dT_i],$$

the direct tax savings, less the payroll recoupment, plus the EITC transfer developed below. The numerator is the beneficiaries' willingness to pay, and it takes one of two forms depending on how customers experience the induced tips.

If tipping comes at customers' expense, the policy is a pure transfer. Per induced tip dollar, the customer pays dT_i and gets nothing in return, the worker keeps the after-tax amount $(1 - \tau_e) dT_i$, and the owner remits the employer share $\tau_e dT_i$; the three positions sum to $-\tau_f dT_i = -g_i$, exactly what the government recoups.²⁹ Summing willingness to pay across customers, workers, and owners therefore reproduces the denominator term by term,

$$N^{\text{expense}}(r) = \sum_i [M_i - g_i + r dT_i] = D(r),$$

and the MVPF is exactly 1 at every r : every dollar the government recoups is paid by some customer, worker, or firm.

If customers tip willingly (the *tipping surplus* case of Section 5), we assume instead that they value the extra tip payment at its full cost: the tip is voluntary, so at the margin the customer is indifferent between the dollar and whatever tipping buys them, and their net willingness to pay is zero. The customer's loss then drops out of the numerator, and in place of $-g_i$ we have the worker's after-tax gain $(1 - \tau_e) dT_i$:

$$N^{\text{willing}}(r) = \sum_i [M_i + (1 - \tau_e) dT_i + r dT_i].$$

The MVPF is $N(r)/D(r)$ for the applicable numerator. At $r = 0$, these yield 1.37 under willing tipping and exactly 1 at customers' expense.

F.2 Payroll recoupment

The payroll tax the government keeps on an induced tip dollar varies with two things: whether the firm is covered by the §45B FICA Tip Credit, and, at covered firms, whether the tip falls above or below the gap between the applicable cash-wage floor and the worker's cash wage. If the firm is outside §45B coverage, no credit is available and $g_i = \tau_f dT_i$. For covered firms, induced tips that fill the gap below the floor earn no credit, so the government keeps both FICA shares on them (τ_f), while induced tips above the gap are creditable: the credit refunds the employer share and the government's net take is the employee share τ_e . We compute creditability at the worker-firm-month

²⁹The sum is $-g_i$ whether or not the tip is creditable under the credit described in the next subsection: the credit refunds the employer share to the owner and reduces g_i by the same amount.

level and annualize. The credit applies to 89% of induced tip dollars. The resulting average is \$16 per worker, a gross effective rate of 8.5%.

F.3 EITC bounds

Workers receive the EITC change without re-optimizing, so it is a first-order transfer to them and an equal cost to the government. Thus the term $r dT_i$ appears in both numerator and denominator. We report $r \in \{0.45, 0, -0.2106\}$: the steepest phase-in (three or more qualifying children), no interaction, and the steepest phase-out (two or more qualifying children), assuming every worker is EITC-eligible and on the sloped region. Both assumptions widen the bound.

These scenarios yield 1.32 and 1.40.

F.4 Social Security offset

The combined FICA rate decomposes into Old-Age and Survivors Insurance (OASI) at 10.6%, Disability Insurance (DI) at 1.8%, and Medicare at 2.9%. Medicare benefits are not earnings-linked, so Medicare taxes carry no marginal benefit obligation to the government and count in g_i in full. OASI taxes do generate an obligation: induced tips remain FICA-covered earnings under the “No Tax on Tips” (NTOT) provision, so each dollar raises the worker’s future retirement benefit and the recouped tax is not pure surplus (Feldstein and Samwick, 1992; Goda et al., 2011). DI benefits are also earnings-linked, but we follow Goda et al. (2011) in treating DI taxes as fully recouped; this omission overstates g_i by at most the DI share of payroll taxes kept (11.8% whether or not the tips are creditable) and therefore overstates the MVPF. The obligation accrues on all induced tips regardless of §45B coverage, because tips are credited to the worker’s earnings record even when the employer claims the credit. We therefore subtract the obligation from g_i directly rather than scaling the OASI dollars the government keeps.

OASI benefits are a progressive function of average indexed monthly earnings (AIME), the average of the worker’s highest 35 years of wage-indexed earnings. The marginal replacement rate $\pi_i \in \{0.90, 0.32, 0.15\}$ depends on where the worker’s AIME falls relative to the formula’s bend points. We assign π_i from annualized 2024 earnings plus induced tips, the post-policy earnings level at which the marginal dollar accrues, using the 2024 bend points: 90% below \$14,088, 32% between \$14,088 and \$84,936, and 15% above; 24% of workers fall in the 90% bracket. This classification assumes each worker’s earnings remain a constant fraction of the national average wage: because past earnings and the bend points are both indexed to average wage growth, the bracket computed from 2024 earnings and 2024 bend points is then the bracket that applies at retirement.

Let σ_i denote the present value of the additional benefits the government becomes obligated to pay per dollar of worker i ’s induced tips. We construct σ_i by walking one dollar through the benefit formula; all flows are in constant 2024 dollars. A dollar earned in 2024 by a worker of age a is credited to the earnings record and revalued by national average wage growth through age 60,

to $(1 + \omega)^{60-a}$, the statutory wage indexation of the AIME formula.³⁰ If 2024 ranks among the worker’s 35 highest-earning years, which holds mechanically for any worker with fewer than 35 years of covered earnings, the revalued dollar enters the 35-year average and raises the annual benefit at the full retirement age of 67 by $\pi_i/35$ times its value: the division by 35 dilutes a single year’s dollar in the career average. For workers with 35 or more higher-earning years the dollar never enters the formula; setting $\sigma_i = 0$ for this case restores the full recoupment and is reported as the alternative. From age 67, the benefit is constant in real terms, and the government pays it in each year the worker survives. Workers may claim as early as 62 or as late as 70, with the monthly benefit scaled down or up by amounts designed to hold its expected present value roughly constant, so fixing the claiming age at 67 is innocuous. Discounting these payments at the real rate δ and weighting by survival gives σ_i , following the survival-weighted construction of Feldstein and Samwick (1992) and Goda et al. (2011):

$$\sigma_i = \frac{\pi_i}{35} (1 + \omega)^{60-a} \sum_{t=67-a}^{\infty} \frac{S_a(t)}{(1 + \delta)^t}$$

where $S_a(t)$ is the probability that a worker of age a survives t more years. We set the representative age $a = 35$, take $S_a(t)$ from the cohort life tables underlying the 2024 Trustees Report (Social Security Administration, 2024), set $\omega = 1.14\%$, the Trustees’ ultimate intermediate real growth rate of the average wage index (Table V.B1 of Social Security Administration, 2024), and set $\delta = 3\%$, the real discount rate of the MVPF framework (Hendren and Sprung-Keyser, 2020).

We subtract $\sigma_i dT_i$ from g_i and add the same accrual to workers’ willingness to pay under willing tipping; valued at cost, the transfer preserves the identity $N^{\text{expense}}(r) = D(r)$, so the floor remains exactly 1. The offset averages \$14.34 per worker (7.4 cents per induced tip dollar) and moves the willing-tipping MVPF from 1.38 to 1.37. The obligation absorbs most of the \$16 gross payroll recoupment per worker: net of future benefits, the government keeps little of the payroll tax it collects on induced tips. Per dollar of OASI tax, the implied obligation is 70 cents, consistent with lifetime benefit-to-tax ratios above 1 for the Social Security Administration’s very-low and low scaled earners (Rose and Burkhalter, 2025) and with net marginal OASI rates near or below zero for low earners (Feldstein and Samwick, 1992). If workers do not perceive the tax-benefit linkage (Liebman et al., 2009), the accrual drops from the numerator only and the MVPF falls to 1.34. Varying the representative age between 25 and 50 and lowering δ to 2.3%, the long-run yield on trust fund assets (Rose and Burkhalter, 2025), moves the MVPF between 1.36 and 1.37.

Three omitted refinements have known directions: spousal and survivor benefits ride on the worker’s record, so their omission overstates the MVPF, while using population rather than earnings-differential mortality and ignoring the income taxation of benefits both overstate σ_i and thus understate the MVPF. These refinements are unlikely to matter for the headline number: they

³⁰The SSA indexes each year’s earnings by growth in the national average wage through the year the worker turns 60, so this is a real-wage adjustment, not an inflation adjustment.

adjust σ_i , and the total effect of σ_i is already small; ignoring the accrual entirely only moves the MVPF from 1.37 to 1.38.

F.5 Incorporating the clawback

Let s be the clawback share from Section 4. Consider one induced tip dollar at a firm that cuts base wages by s . The worker loses s in wages but no longer remits employee FICA or income tax on them, so the worker's after-tax loss is $(1 - \tau_e - m_i)s$, where m_i is the worker's marginal income-tax rate; the tax calculations underlying Section 2 supply m_i for each worker, and it is zero for the 26.4% of workers below the standard deduction. The owner keeps the clawed-back wages and no longer remits employer FICA on them, a gain of $(1 + \tau_e)s$. The government loses both FICA shares and the forgone income tax, $(\tau_f + m_i)s$: the response's fiscal externality. The three positions sum to zero, as a transfer must. Net new earned income per tip dollar is $1 - s$, so the EITC term and the Social Security accrual both scale by $(1 - s)$. The netting enters through g_i , so all MVPF figures reported here are net of this offset.

The clawback's effect depends on the social welfare weight placed on firm owners: with equal weights, both workers' and owners' willingness to pay count in the numerator; with owners' weight set to zero, only workers' does (Section 5).

With the clawback, the denominator and the two willing-tipping numerators are

$$\begin{aligned} D_s(r) &= \sum_i [M_i - g_i + (\tau_f + m_i)s dT_i + r(1 - s) dT_i] \\ N_s^{\text{equal}}(r) &= \sum_i [M_i + (1 - \tau_e)(1 - s) dT_i + m_i s dT_i + (1 + \tau_e)s dT_i + r(1 - s) dT_i] \\ N_s^{\text{workers}}(r) &= \sum_i [M_i + (1 - \tau_e)(1 - s) dT_i + m_i s dT_i + r(1 - s) dT_i] \end{aligned}$$

Relative to the baseline, $N_s^{\text{equal}}(r)$ and $D_s(r)$ move by the same $\sum_i (\tau_f + m_i - r)s dT_i$, which is why the equal-weights MVPF is essentially unchanged; $N_s^{\text{workers}}(r)$ drops the owners' term, which is why that MVPF falls. The identity $N_s^{\text{expense}}(r) = D_s(r)$ is preserved, so the floor remains exactly 1.

These yield 1.36 with equal weights and 1.25 (1.20 to 1.30 across the clawback's 95% confidence interval) with owners' weight set to zero.

F.6 Unmodeled channels

The calculation omits six channels; we sign each and size the ones we can.

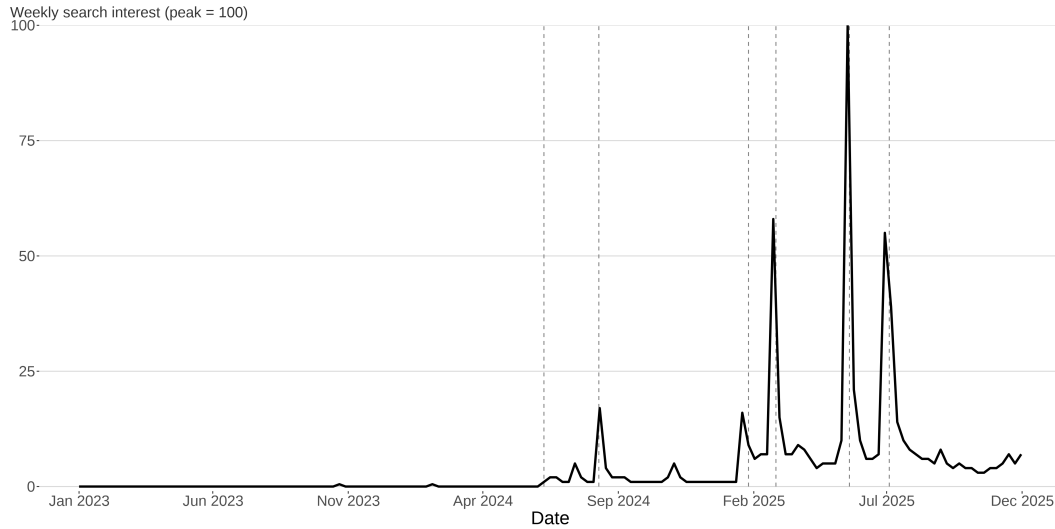
1. Higher effective hourly pay could raise workers' hours. The taxes on those extra earnings are a fiscal externality we omit, so the omission understates the MVPF; the workers' own gain from re-optimizing is second-order by the envelope theorem. In the incidence sample, hours

did not rise with the dose: monthly hours at adopting firms show no significant response in the event window, with point estimates that are small and negative though noisily estimated. This channel is therefore unlikely to be understating the MVPF.

2. Owners' clawback gain is pre-tax. Taxes on firm profits (21% for C corporations, owners' individual rates net of the qualified business income deduction for pass-throughs) would shift part of that gain to the government; willingness to pay and net cost fall together, so the equal-weights figure of 1.36 is essentially unchanged, while the workers-only figure of 1.25 rises because the added revenue lowers net cost without touching workers' willingness to pay. The shift is at most $\tau_\pi(1 + \tau_e)s dT_i$ per worker, roughly \$12 at the C-corporation tax rate of 21%. This raises the workers-only figure by roughly \$12 over the per-worker net cost, a change of a few percent, and the equal-weights figure by that amount scaled by the MVPF's excess over 1, a change an order of magnitude smaller.
3. Customers' tipped dollars might otherwise have been spent and taxed. Ignoring this counterfactual revenue overstates the recoupment and hence the MVPF. The federal tax content of a marginal consumption dollar is small. Only the small profit share of displaced revenue faces the corporate rate and only the labor share generates payroll and income taxes, while sales taxes are state and local. The omission is therefore a few cents per displaced dollar at most, scaled further by whatever fraction of tips displaces other spending.
4. The employer share of FICA on non-covered induced tips is collected in g_i but debited from no one's willingness to pay. A strict accounting would subtract \$1.64 per worker from owners, slightly lowering the willing-tipping MVPF.
5. The clawback transfer is counted in full for every adopting owner, but adoption may carry costs we do not measure, and the marginal adopter was left indifferent by them (Finkelstein and Hendren, 2020). To that extent we overstate owner surplus by the aggregate of those adjustment costs. Two features of the setting suggest these costs are near zero. The first is that enabling tipping requires only a settings change in the point-of-sale software. The second is that 45% of the 156 adopters in our incidence sample enable tipping while collecting essentially no tips in the three subsequent months, which is consistent with a trivial cost of turning the feature on.
6. The calculation is federal-only: excluding state income taxes overstates workers' net gain from induced tips under willing tipping, while the floor is unaffected. In addition, roughly half of income-taxing states extend the deduction through federal conformity or their own legislation (PBS NewsHour, 2026), which shrinks this overstatement.

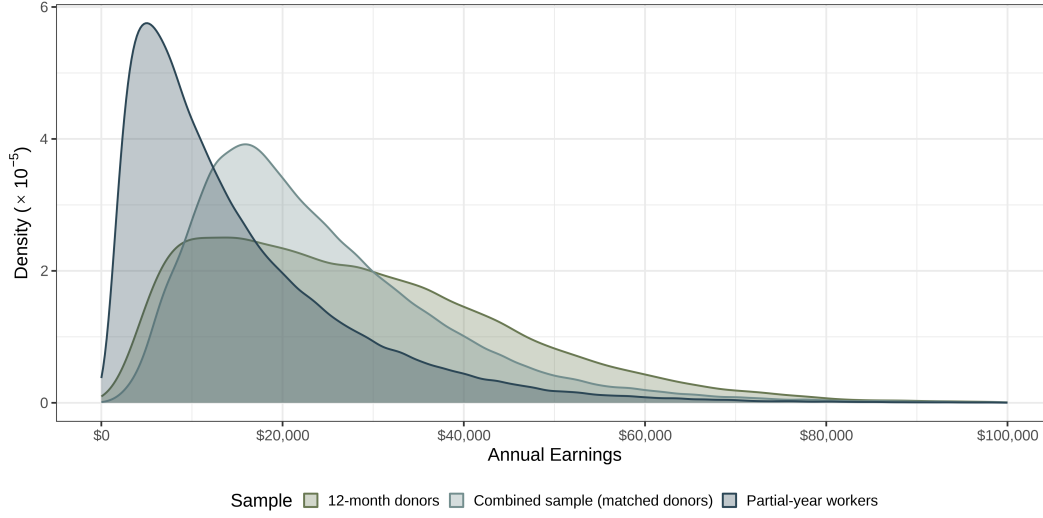
Appendix Figures

Figure A1: Google Search Interest for “No Tax on Tips”



Notes: This figure presents data from Google Trends for the search term “no tax on tips” in the United States from January 1st, 2023 through October 20th, 2025. Values represent search interest relative to the peak during this period. A value of 100 indicates maximum popularity and 0 indicates insufficient search volume. The first and second gray vertical lines denote the weeks in which Donald Trump and Kamala Harris made their initial campaign promises. The third marks President Trump’s remarks at a campaign rally regarding the tax proposal. The fourth denotes when the House adopted a budget resolution to advance the “One Big Beautiful Bill” Act (OBBBA), renewing interest in the proposal. The last two dates mark Senate passage of the standalone “No Tax on Tips” Act and the signing of the OBBBA into law on July 4th, 2025. Source: <https://trends.google.com/trends/explore?date=2023-01-01%202025-10-20&geo=US&q=no%20tax%20on%20tips&hl=en>

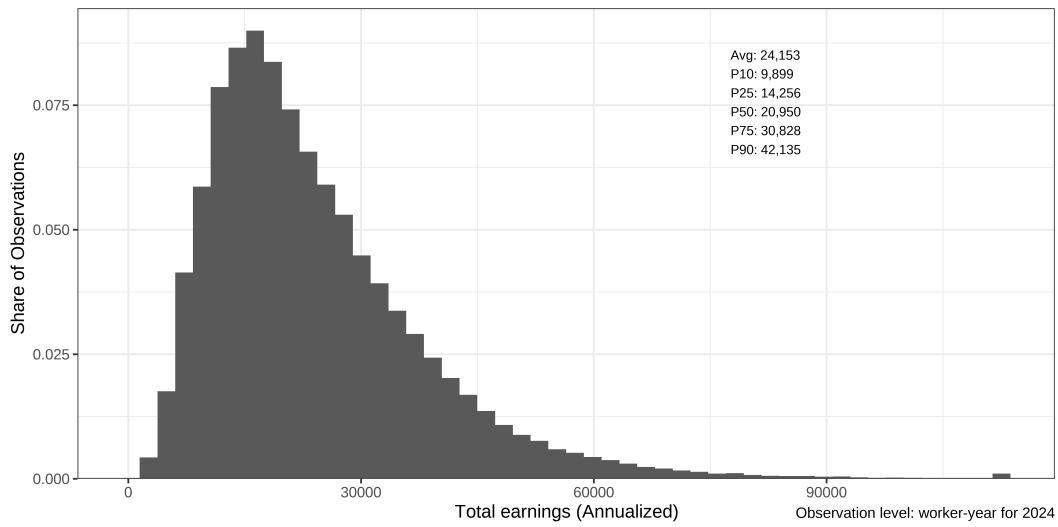
Figure A2: Annual Earnings Distribution by Sample Construction Method



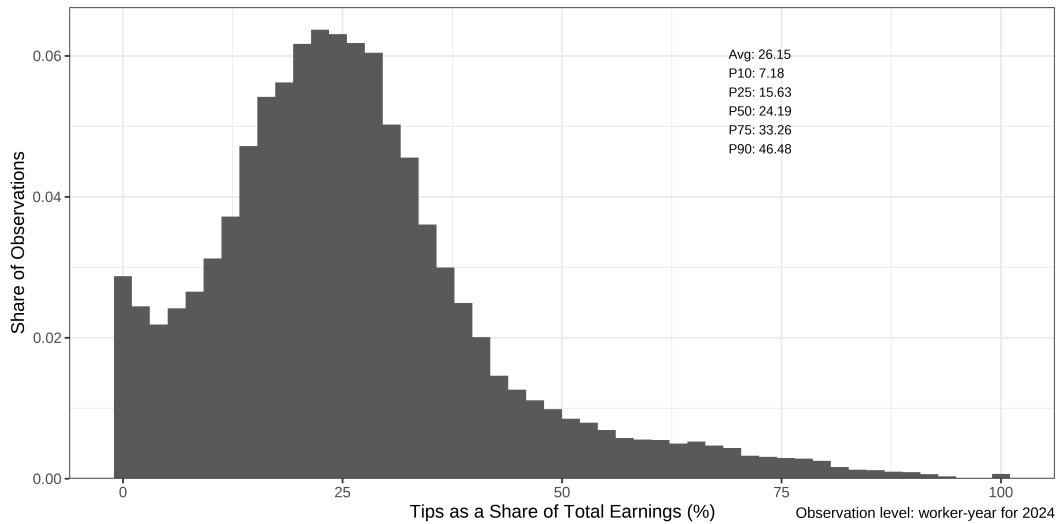
Notes: This figure presents kernel density estimates of annual earnings for tipped workers in 2024 across three sample construction approaches. The “12-month donors” sample comprises workers observed for the full calendar year whose earnings profiles are used in the donor matching procedure. The “partial-year workers” sample includes workers observed for fewer than 12 months. The “combined sample” integrates both groups using donor-matching to construct annual earnings estimates for partial-year workers. For each partial-year worker-firm-month observation, we identify donors from the 12-month sample who share the same ZIP3, industry classification, and sub-industry, selecting the 10 closest matches based on Euclidean distance across standardized worker and firm characteristics (monthly earnings, hours, base wage, tip earnings, firm revenue, and firm size). Only matches with monthly distance less than 3 across all observed months are retained. The combined sample weights lower-earning workers who appear for part of the year, providing a representative distribution of annual earnings for the range of tipped workers in our data.

Figure A3: Annual Earnings and Tip Share Distribution for Tipped Workers

(a) Annual Earnings

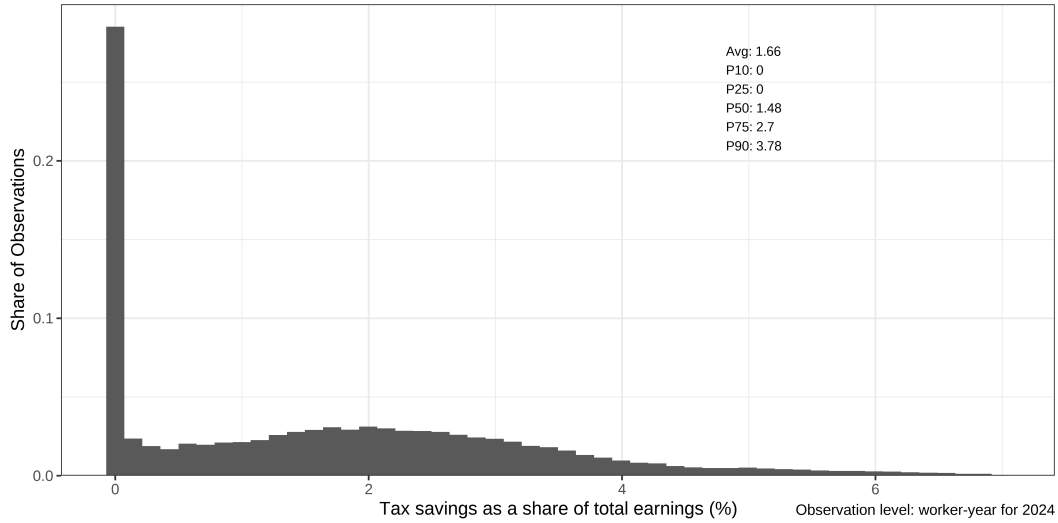


(b) Share of Earnings from Tips



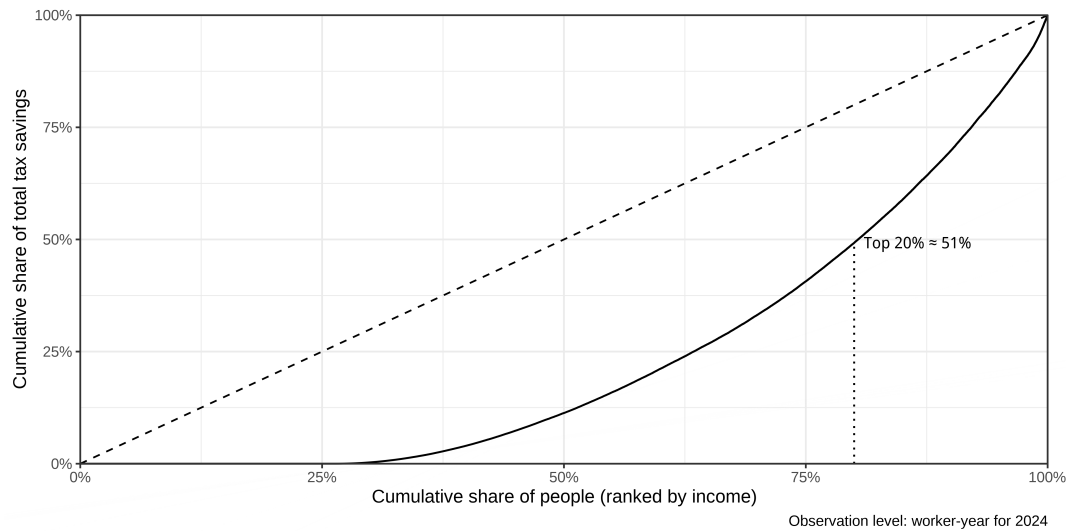
Notes: This figure presents the distribution of annual earnings and tip income composition for tipped workers in 2024. Panel (a) displays the histogram of total annual earnings for the combined sample. The distribution exhibits substantial right skew, with median earnings of \$20,950 and the 90th percentile at \$42,135, indicating concentration at the lower end of the income distribution. Panel (b) shows the share of annual earnings derived from tips. The mean tip share is 26%, though 2.8% of workers derive less than one percent of their income from tips despite having some tip earnings during the year. The combined sample integrates workers observed for the full calendar year with partial-year workers whose annual earnings are estimated via donor-matching as described in Appendix C.1.

Figure A4: Federal Income Tax Savings as Percentage of Annual Earnings



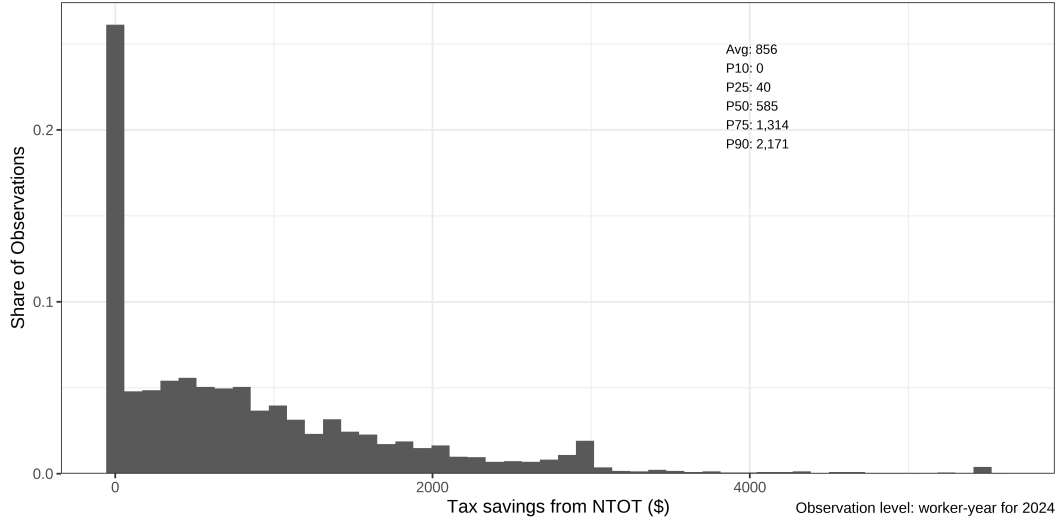
Notes: This figure displays the distribution of tax savings as a share of total annual earnings for tipped workers under the NTOT provision. The mean tax reduction equals 1.7% of annual earnings, though this masks considerable variation. Workers earning below the standard deduction threshold receive zero benefit (0% of earnings), while those in higher tax brackets with substantial tip income receive proportionally larger savings. Conditional on any tax savings, the average share of earnings saved is 2.25%. Tax savings calculations follow the methodology described in Figure 1 notes.

Figure A5: Cumulative Distribution of Tax Savings by Worker Income Percentile



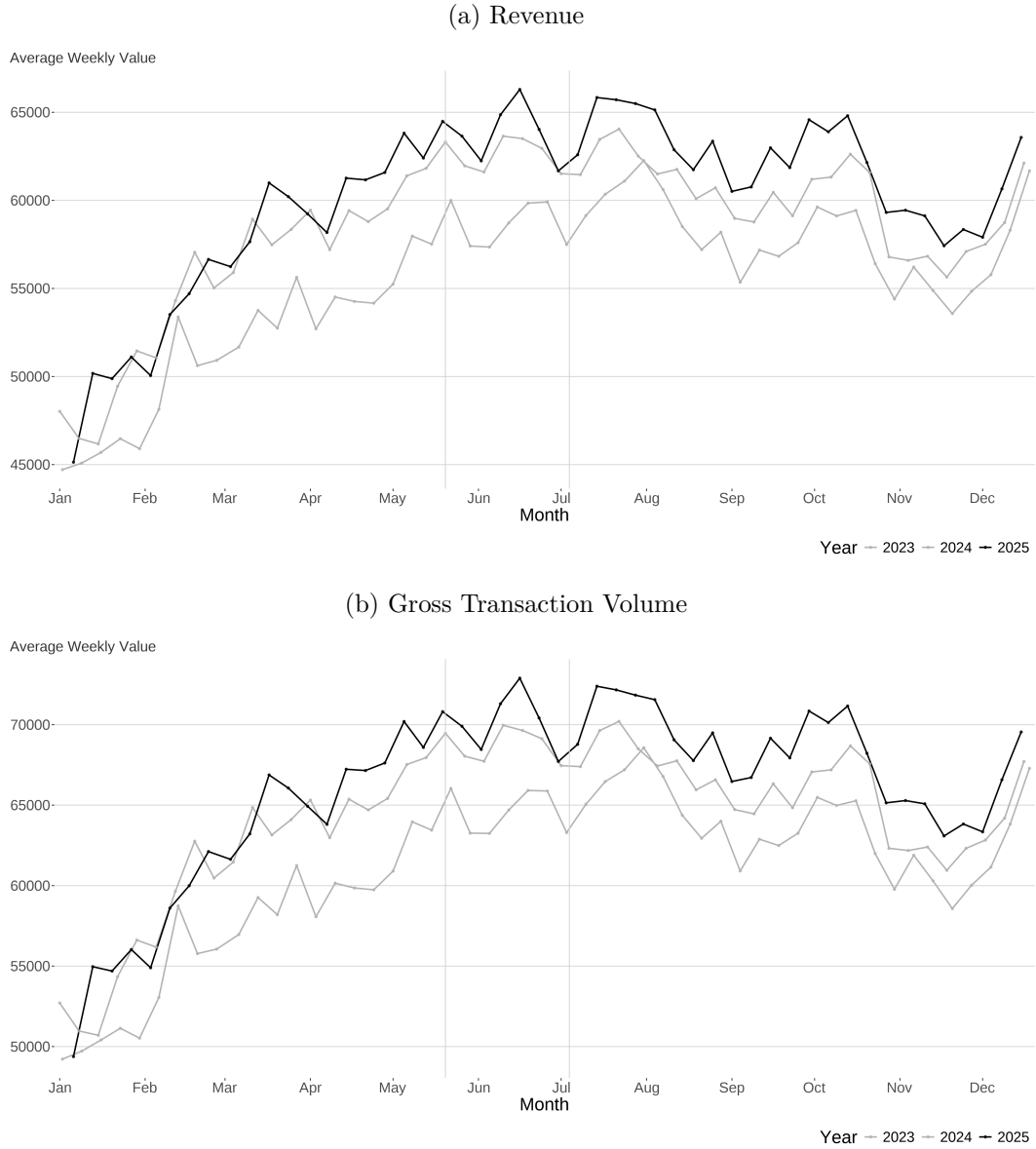
Notes: This figure plots the concentration curve for NTOT tax savings, showing the cumulative share of total tax benefits captured by cumulative shares of workers ranked by 2024 annual earnings. The 45-degree line represents perfect equality where each percentile of workers receives an equivalent percentile of total tax savings. Deviation above this line indicates concentration among higher earners. The top 20% of tipped workers capture 51% of aggregate tax savings, while the bottom 50% receive only 11.3% of total benefits. This concentration reflects two mechanisms: (1) workers earning below the standard deduction threshold receive zero benefit despite representing 26.4% of the sample, and (2) the progressive federal tax structure means workers in higher brackets save more per dollar of tip income deducted.

Figure A6: Distribution of Tax Savings from NTOT: Validated Sample



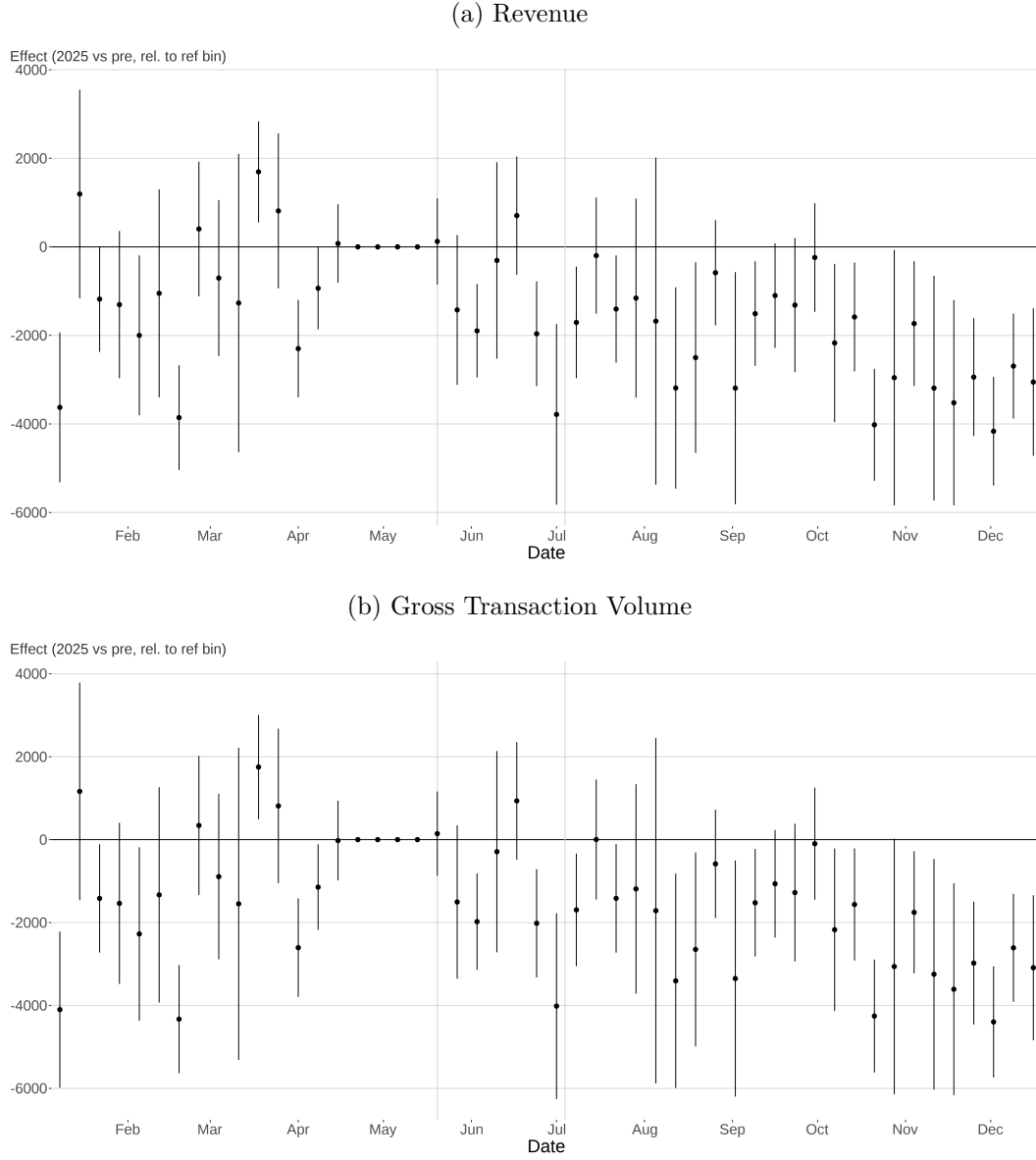
Notes: This figure presents the distribution of estimated tax savings from the “No Tax on Tips” provision for workers in the validated sample, using the sample construction and tax calculation methodology described in Appendix C.3.1. We assume single filing status, standard deduction only, and no income sources beyond observed earnings. Average tax savings are \$856 per worker, compared to \$531 in the main sample. The higher average reflects the validated sample’s higher earnings rather than differential tip intensity, as evidenced by similar tip share distributions across samples. The validated sample exhibits different patterns from the main sample: 18.5% of workers receive zero tax savings because total earnings fall below the standard deduction threshold, compared to 26.4% in the main sample. This difference reflects the validated sample’s construction, which relies on higher-earning firms that appear in both data sources. Values of \$3,000 and \$5,500 reflect workers who deduct the full \$25,000 tip income deduction at marginal tax rates of 12% and 22%, respectively. Among workers with positive savings (81.5% of the validated sample), average savings are \$1,051 (median \$788), representing approximately 2.71% of annual earnings for beneficiaries.

Figure A7: Weekly Sales Patterns by Calendar Year, 2023–2025



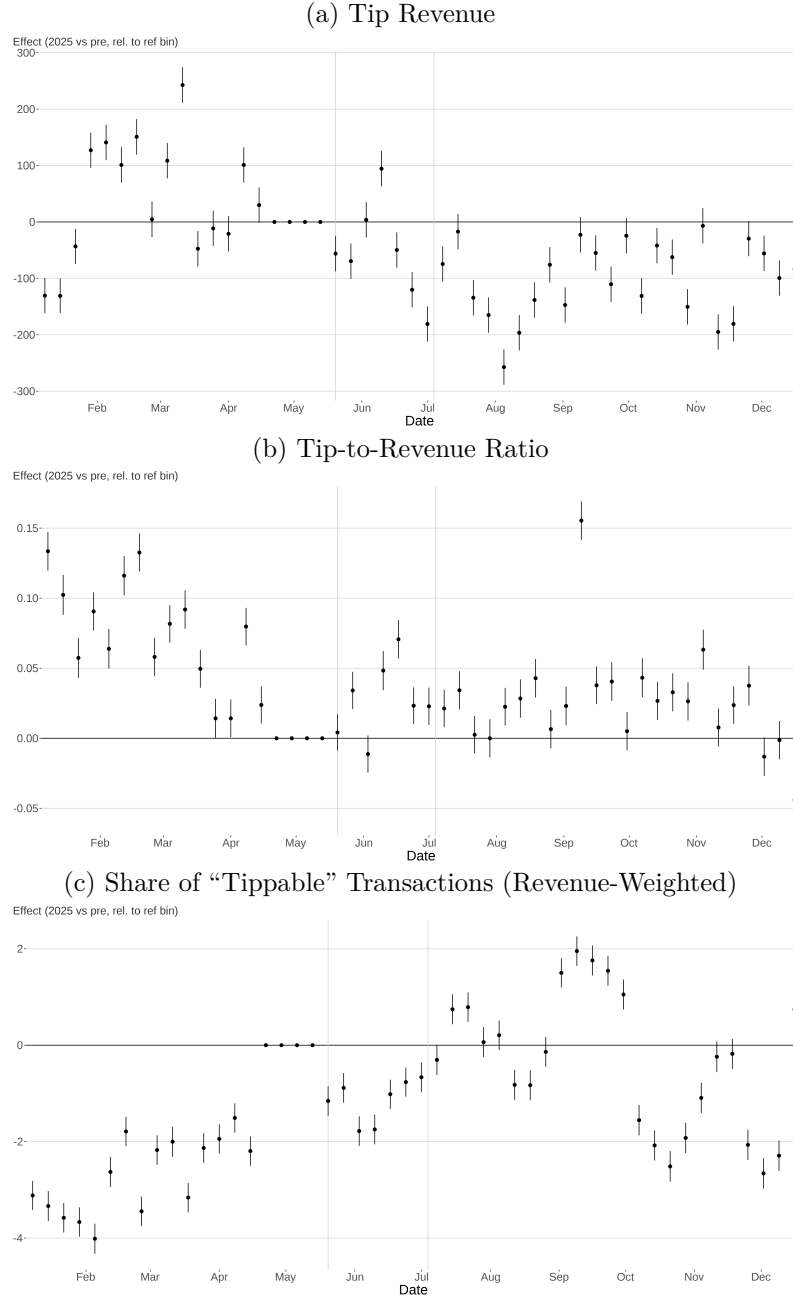
Notes: This figure presents weekly time series of revenue and gross transaction volume (revenue plus sales tax and tips) to assess whether the tipping response documented in Figure 2 reflects compositional changes in transaction volumes rather than pure tipping behavior. Panel (a) displays average revenue per firm-week. Panel (b) shows average gross transaction volume. Both panels present 2025 observations in black and 2023–2024 in gray. All measures represent unweighted simple averages across firms in each week. The last two weeks of each calendar year are excluded. Vertical lines mark May 20th (Senate NTOT passage) and July 4th (OBBBA signing). The absence of systematic divergence in 2025 from historical patterns indicates that increased tip revenue documented in the main analysis reflects genuine changes in tipping behavior rather than increased transaction volumes or higher bill sizes.

Figure A8: Event Study Estimates for Revenue and Gross Transaction Volume



Notes: This figure presents event study estimates for revenue and gross transaction volume (revenue plus sales tax and tips) corresponding to the time series displayed in Figure A7. Under Assumption 1, each panel plots coefficients β_ℓ from Equation (2), which compares weekly outcomes in 2025 to their calendar-time counterparts in 2023–2024, relative to the four weeks immediately preceding May 20th. The last two weeks of each calendar year are excluded. Panel (a) displays average revenue per firm-week. Panel (b) shows average gross transaction volume. Vertical lines mark May 20th (Senate NTOT passage) and July 4th (OBBA signing). Point estimates reveal no statistically significant changes in either measure following NTOT passage, though confidence intervals allow for moderate increases or decreases. These null findings support the interpretation that the tip revenue increases documented in Figure 3b stem from increasing tipping availability and higher tip-to-revenue ratios rather than increased customer traffic or larger purchases. Standard errors follow Driscoll and Kraay (1998) with a lag length of 8 weeks.

Figure A9: Placebo Test: Event Study Using 2024 as Treatment Year



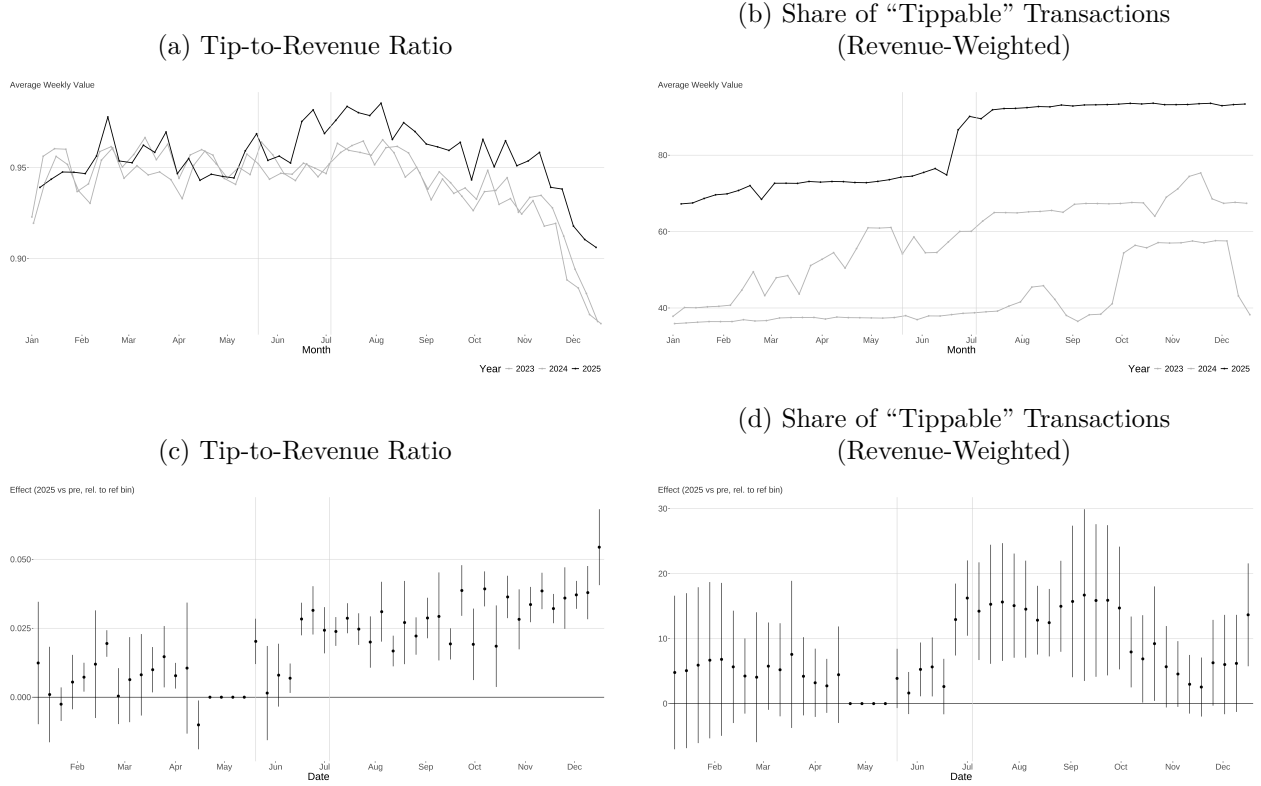
Notes: This figure presents placebo tests using 2024 as a pseudo-treatment year to assess whether the behavioral responses documented in Section 3.3 reflect genuine policy effects or spurious calendar-year patterns. We estimate Equation (2) treating May 20th, 2024, as the “treatment” date and using 2023 as the control year. The last two weeks of each calendar year are excluded, as in the main specification. Panel (a) displays tip revenue, Panel (b) shows tip-to-revenue ratios, and Panel (c) presents the revenue-weighted share of tippable transactions. Vertical lines mark the analogous May 20th and July 4th dates in 2024. No panel shows systematic positive shifts following the pseudo-treatment date, in contrast to the 2025 results (Figures 3, A10, A11, A12, and 4), supporting Assumption 1. Standard errors follow Driscoll and Kraay (1998) with a lag length of 8 weeks.

Figure A10: Tipping Behavior Changes in Food and Beverage Firms



Notes: This figure replicates the analysis from Figures 2 and 3 for the subsample of food and beverage firms. For this analysis, we use the full firm roster rather than the top-decile analysis sample used in the overall sample figures. Panels (a) and (b) present weekly time series for 2023–2025, with 2025 in black and prior years in gray. Panels (c) and (d) display event study estimates of β_ℓ from Equation (2), with the four weeks preceding May 20th as the omitted baseline. The last two weeks of each calendar year are excluded. Vertical lines mark May 20th (Senate NTOT passage) and July 4th (OBBBA signing). Results show a 2 percentage point increase in the share of tippable transactions and a 0.10 percentage point increase in tip-to-revenue ratio following policy passage, directionally consistent with the overall sample findings. Standard errors follow Driscoll and Kraay (1998) with a lag length of 8 weeks.

Figure A11: Tipping Behavior Changes in Retail Firms



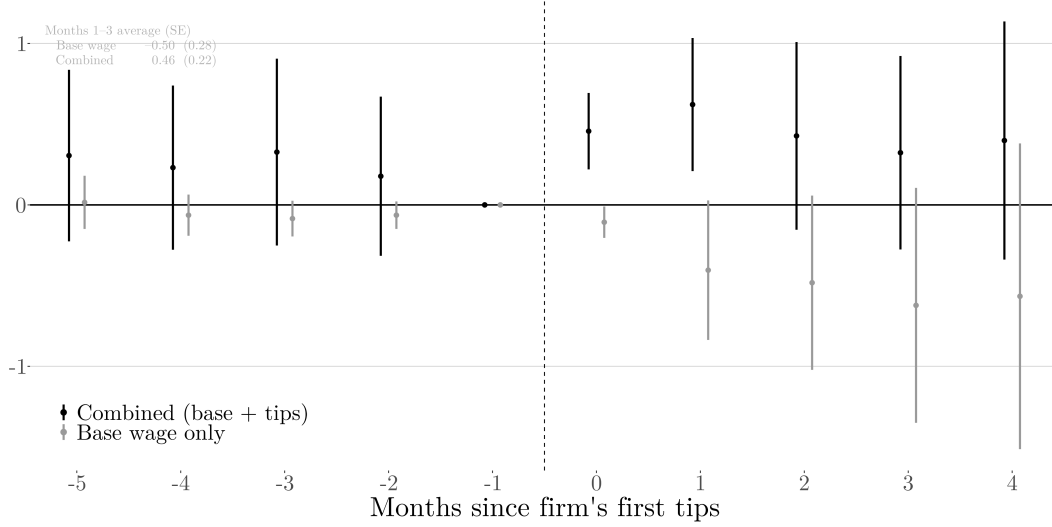
Notes: This figure replicates the analysis from Figures 2 and 3 for the subsample of retail firms. For this analysis, we use the full firm roster rather than the top-decile analysis sample used in the overall sample figures. Panels (a) and (b) present weekly time series for 2023–2025, with 2025 in black and prior years in gray. Panels (c) and (d) display event study estimates of β_ℓ from Equation (2), with the four weeks preceding May 20th as the omitted baseline. The last two weeks of each calendar year are excluded. Vertical lines mark May 20th (Senate NTOT passage) and July 4th (OBBBA signing). Results show a 15 percentage point increase in the share of tippable transactions and a 0.02 percentage point increase in the tip-to-revenue ratio following policy passage. Standard errors follow Driscoll and Kraay (1998) with a lag length of 8 weeks.

Figure A12: Tipping Behavior Changes in Firms with 95–100% Initial Tipping Share



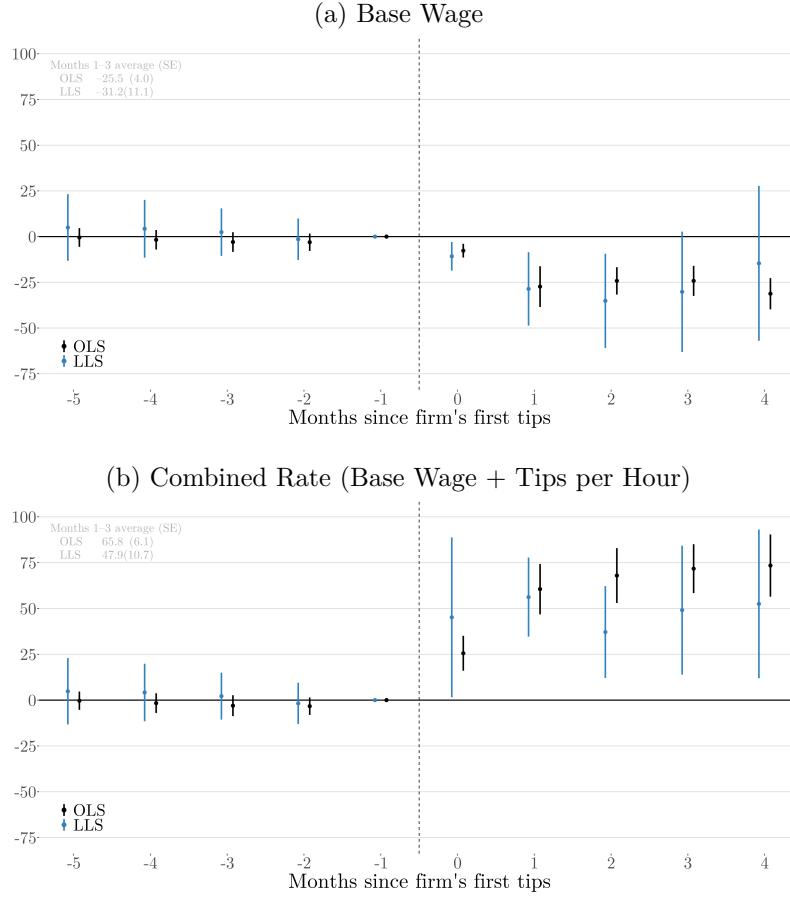
Notes: This figure examines firm-years classified as having 95–100% revenue-weighted share of tippable card transactions in the first four weeks of each year. Firms can move between classifications across years based on their initial four-week tipping intensity. For this analysis, we use the full firm roster rather than the top-decile analysis sample used in the overall sample figures. Panels (a) and (b) show weekly time series for 2023–2025, with 2025 in black and prior years in gray. Vertical lines mark May 20th (Senate NTOT passage) and July 4th (OBBBA signing). Panels (c) and (d) present event study estimates from Equation (2), with the four weeks preceding May 20th as the omitted baseline, excluding the first four weeks of each year (as these weeks define the subsample) and the last two weeks of each year (to avoid holiday-driven volatility). Despite starting at a 98.6% share of “tippable” transactions over the 2023–2024 pre-period, these firms maintained elevated levels throughout 2025 while previous years showed steady declines. However, when we estimate Equation (2), which exploits within-firm variation, we see no statistically significant increase in the revenue-weighted share of “tippable” transactions. Tip revenue also shows no significant change. Standard errors follow Driscoll and Kraay (1998) with a lag length of 8 weeks.

Figure A13: Unconditional Adoption Event Study: Base-Wage and Combined Hourly Rates



Notes: This figure presents event-study estimates of the effect of a firm's adopting tipping on the base-wage rate (gray) and the combined hourly rate (base wage plus tips per hour; black), without the dose interaction used in Figure 5. Each series plots the adoption-month dummies δ_k from Equation (5), measured relative to the month before adoption. The sample is the same as in Figure 5: 689 workers at 156 firms that adopted tipping between March and August 2025, with not-yet-treated firms as controls. Post-treatment averages are taken over months 1–3; month 0 is the partial adoption month, and month 4 is displayed but excluded from the average. The base-wage rate is near zero with tight confidence intervals before adoption (standard errors of 0.04–0.08); afterward, point estimates decline but standard errors widen to 0.22–0.48, rendering the individual coefficients insignificant. The combined rate has wide confidence intervals throughout the pre-period (standard errors of 0.25–0.29); they narrow at adoption (0.12 at month 0) before returning to pre-period levels by month 2. This pattern reflects heterogeneity in the treatment effect across firms with different tipping intensities: the unconditional specification averages over the dose distribution, inflating the residual variance. The dose-response design in Section 4 addresses this heterogeneity by conditioning on the firm's tipping intensity. Standard errors are clustered at the employer level.

Figure A14: Local Least-Squares versus OLS Estimates of the Dose Response



Notes: This figure compares the event-study path of the dose response under the local least-squares (LLS) estimator of Balla-Elliott (2026) (blue) with the corresponding OLS estimates (black). Panel (a) presents the base-wage rate and Panel (b) the combined rate (base wage plus tips per hour); coefficients are in cents of hourly pay per dollar per hour of dose, measured relative to the month before adoption. At each event month, the LLS estimate averages the dose-specific responses described in Appendix E.2.2, estimated with a bandwidth of \$0.40 per hour. Each local regression compares 58 treated firms with 64 not-yet-treated firms; controls are later adopters observed before their own adoption and, like the treated firms, are restricted to those with positive doses. The OLS series likewise drops zero-dose firms, so the time effects are identified from positive-dose firms only, and the estimates therefore differ slightly from the main-text specification (Section 4), which retains zero-dose adopters with a separate adoption effect. Post-treatment averages are taken over months 1–3; month 0 is the partial adoption month, and month 4 is displayed but excluded from the average. Over this window, the base-wage rate declines by 31.2 cents under LLS (SE 11.1) versus 25.5 cents (SE 4.0) under OLS; the combined rate rises by 47.9 cents (SE 10.7) under LLS versus 65.8 cents (SE 6.1) under OLS. The implied LLS clawback is 39 cents per dollar of induced tips (SE 13.3), compared with 28 cents (SE 3.7) in the main specification. LLS confidence intervals use a normal approximation with standard errors from a firm-level bootstrap with 5,000 replications; OLS standard errors are clustered on the employer.

Appendix Tables

Table A1: Descriptive Statistics for Workers and Firms in the Direct Effect Sample

	Mean	SD	P25	P50	P75
Panel A: Job Characteristics					
Monthly Earnings (\$)	1,970	1,465	900.91	1,646	2,690
Tip Earnings (\$)	506.76	599.89	121.00	328.91	681.00
Hourly Wage Rate (\$)	21.77	10.66	16.38	20.11	24.81
Monthly Hours Worked	89.91	52.06	48.15	82.85	125.57
Tenure (months)	11.83	0.99	12.00	12.00	12.00
% Full Time (>130 hrs/mo)	32.4				
Number of Workers	172,169				
Panel B: Firm Characteristics					
Monthly Revenue (\$)	52,391	71,428	18,229	34,488	63,074
Revenue per Hour (\$)	84.08	110.88	44.58	64.57	94.15
Average Workers per Firm	9.88	8.95	4.00	7.00	12.00
Within-Firm Wage SD (\$)	2.81	6.73	0.00	1.40	3.36
Mean Firm Hourly Wage (\$)	23.33	13.91	17.00	21.05	26.35
Number of Firms	13,294				

Notes: This table provides summary statistics on workers and firms in our combined sample (as described in Appendix C.1). Panel A reports job characteristics calculated across worker-firm-month observations in 2024. Monthly earnings are calculated as total base wages plus any overtime or double-time earnings plus tips. Hourly wage rate is computed as total earnings in a given month divided by the total number of hours worked. Tenure is calculated as months in the analytic sample per job. Full-time status is defined as working more than 130 hours per month; the reported share represents the unweighted proportion of worker-firm-month observations exceeding this threshold. Panel B reports firm characteristics calculated through a two-step procedure: first, firm-month statistics are computed by aggregating worker-firm-month observations; second, summary statistics are calculated across these firm-month observations. Revenue per hour is total monthly revenue divided by total hours worked by all employees, for the firms where we observe both payroll and revenue data. Within-firm wage standard deviation (SD) measures the standard deviation of hourly earnings across workers within each firm-month.

Table A2: Firm Size Distribution in the Direct Effect Sample

Variable	Worker-weighted Sample	Firm Sample
<i>Firm Size</i>		
1 - 4	9.4	25.7
5 - 9	28.5	38.0
10 - 14	23.1	18.7
15 - 19	13.8	8.2
20 - 24	8.2	3.9
25 - 49	13.6	4.9
50+	3.3	0.6

Notes: This table presents the firm size distribution for the direct effect estimation sample. The worker-weighted sample column reports the percentage of workers, while the firm sample column reports the percentage of firms. Firm size reflects the median number of workers per firm-month in 2024. Small firms (1–4 workers) constitute 25.7% of firms but employ 9.4% of workers, while larger firms (25+ workers) represent 5.5% of firms but employ 17.0% of workers. The sample includes 13,294 firms and 172,169 workers.

Table A3: Transaction-Level Descriptive Statistics by Industry and Tipping Intensity

Variable	Overall	Food & Bev	Retail	Near One	Near Zero
Panel A: Weekly averages					
Firms	5,657	41,296	9,897	42,544	12,781
Weeks per Firm	155	154	154	127	102
Pre-period weekly transactions	2,361	591	416	547	502
Revenue	\$57,910	\$13,560	\$19,781	\$13,171	\$22,284
Tip Revenue	\$3,104	\$1,066	\$150	\$1,113	\$44
Number of Transactions	2,858	752	555	704	665
Number of Tippable Transactions	2,034	546	262	552	127
Number of Tipped Transactions	1,058	299	34	305	7
Panel B: Transaction-weighted weekly volumes					
Share of “Tippable” Transactions (Wtd)	83.9%	90.8%	61.6%	98.6%	24.8%
Bill Size	\$20.27	\$18.03	\$35.61	\$18.70	\$33.51
Bill Size (Tippable Transactions)	\$18.36	\$18.50	\$35.96	\$19.62	\$30.63
Bill Size (Tipped Transactions)	\$17.38	\$19.85	\$30.71	\$19.97	\$36.57
Tip per Transaction	\$1.09	\$1.42	\$0.27	\$1.58	\$0.07
Tip per Tippable Transaction	\$1.52	\$1.94	\$0.57	\$2.01	\$0.34
Tip per Tipped Transaction	\$2.93	\$3.57	\$4.45	\$3.65	\$6.49
Tip to Revenue Percent	6.6%	7.5%	1.0%	8.4%	0.2%
Tip Percent (Pre-Tax; Tippable)	9.1%	10.0%	2.0%	10.2%	0.9%
Tip Percent (Pre-Tax; Tipped)	17.2%	17.9%	15.9%	18.1%	16.9%

Notes: This table presents transaction-level descriptive statistics for the behavioral response samples, disaggregated by industry classification and baseline tipping intensity. Panel A reports unweighted weekly firm-level averages. Panel B weights observations by the relevant transaction count for each measure: overall statistics are weighted by the total transactions, tippable-specific measures by the number of tippable transactions, and tipped-specific measures by number of tipped transactions. This weighting reflects the average transaction in the sample. The “Overall” column further restricts the firm roster to those in the top decile of average weekly card transactions during 2023–2024 (roughly 1,000+ per week); the other columns use the full firm roster. The “Food & Bev” and “Retail” subsamples are defined by industry. “Near Zero” and “Near One” subsamples include firm-years with the revenue-weighted share of tippable transactions in the first four weeks of each year between 0–5% and 95–100%, respectively. Transaction counts distinguish between all transactions, “tippable” transactions (those with tipping functionality enabled), and “tipped” transactions (those where customers left tips). Tip percentages are calculated relative to pre-tax, pre-tip subtotals.

Table A4: Descriptive Statistics for Workers and Firms in the Validated Sample

	Mean	SD	P25	P50	P75
Panel A: Job Characteristics					
Monthly Earnings (\$)	2,575	1,842	1,299	2,294	3,477
Tip Earnings (\$)	721.85	780.40	206.00	515.62	990.69
Hourly Wage Rate (\$)	23.94	10.87	17.89	22.26	27.54
Monthly Hours Worked	106.42	53.14	64.00	105.91	146.93
Tenure (months)	11.88	0.85	12.00	12.00	12.00
% Full Time (>130 hrs/mo)	35.0				
Number of Workers	3,028				
Panel B: Firm Characteristics					
Monthly Revenue (\$)	76,453	86,834	30,217	52,490	91,502
Revenue per Hour (\$)	82.75	114.47	49.68	66.21	88.62
Average Workers per Firm	12.53	10.51	6.00	10.00	16.00
Within-Firm Wage SD (\$)	4.42	8.09	1.53	2.89	5.14
Mean Firm Hourly Wage (\$)	23.52	9.51	17.52	22.16	27.17
Number of Firms	793				

Notes: This table presents descriptive statistics for the validated sample, comprising workers observed for all 12 months of 2024 at firms that pass two firm-month-level validation criteria: plausible earnings-to-revenue ratios and correspondence between transaction-level tip collection and payroll tip distribution, as described in Appendix C.3.1. Panel A reports job characteristics calculated across worker-firm-month observations in 2024. Monthly earnings are calculated as total base wages plus any overtime or double-time earnings plus tips. Hourly wage rate is computed as total earnings in a given month divided by the total number of hours worked. Tenure is calculated as months in the analytic sample per job. Full-time status is defined as working more than 130 hours per month; the reported share represents the unweighted proportion of worker-firm-month observations exceeding this threshold. Panel B reports firm characteristics calculated through a two-step procedure: first, firm-month statistics are computed by aggregating worker-firm-month observations; second, summary statistics are calculated across these firm-month observations. Revenue per hour is total monthly revenue divided by total hours worked by all employees, for the firms where we observe both payroll and revenue data. Within-firm wage SD measures the standard deviation of hourly earnings across workers within each firm-month. Workers in the validated sample exhibit higher monthly per-firm earnings (\$2,575 vs. \$1,970 in the main sample) and higher hours (106 vs. 90). Firms in the sample are larger (12.5 vs. 9.9 workers) and have roughly 50% higher mean monthly revenue (\$76,000 vs. \$52,000), while exhibiting similar revenue per hour worked (\$83 vs. \$84). These statistics reflect selection by firms into appearing in both the point-of-sale and payroll records.